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RESEARCH

GraphINR-IDS: Graph-Topology-Aware implicit neural reconstruction for zero-day intrusion detection in encrypted network traffic

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Abstract

Payload encryption has shifted network intrusion detection towards flow-metadata analysis, yet most metadata-based detectors represent traffic through fixed feature windows, token sequences or periodic graph snapshots, discretising traffic before learning and binding the detector to a granularity selected in advance. GraphINR-IDS instead models benign flow behaviour as a continuous reconstruction function over source–destination topology, capture time and observable metadata. Spectral coordinates of the graph Laplacian encode communication position, multi-frequency sinusoids encode capture time, and a SIREN network reconstructs each flow from its own coordinate, with anomaly decisions following from standardised feature-wise residuals. Fitting, standardisation and calibration use benign flows exclusively, so the attack families met at test time are absent from fitting by construction. Twelve one-class detectors were compared on UNSW-NB15 and CIC-IDS2017 under one random and two chronological protocols. GraphINR-IDS reached 99.06 ± 0.12 % F1 at a 5.13 % false-positive rate, with the strongest threshold-free ranking of the twelve (AUROC 99.66 %, AUPRC 99.88 %). A fixed-capacity LSTM comparator lost roughly 55 to 67 percentage points of F1 as its window grew from 8 to 64 flows, whereas the continuous formulation requires no equivalent selection. Separate ablations attributed 1.15 percentage points of F1 to the spectral coordinate and 0.31 to hierarchical frequency decomposition, with no measurable contribution from graph-conditioned modulation on the static graphs evaluated. Chronological transfer raised realised false-positive rates for every detector, and re-estimating feature-wise residual statistics alongside the threshold recovered substantially more performance than threshold-only recalibration. The evidence supports continuous graph-temporal reconstruction as a competitive benign-only formulation and identifies calibration drift as a principal limitation under the evaluated protocols. Direct validation on operational encrypted traffic and evolving topologies remains necessary.

Keywords Implicit neural representations · Intrusion detection · Encrypted traffic analysis · Graph spectral encoding · Anomaly detection · Zero-day attacks · SIREN · UNSW-NB15

1 Introduction

Transport-layer encryption has changed what an intrusion detector can observe. TLS 1.3 and QUIC now carry most internet-facing traffic, and the encrypted share of enterprise traffic continues to grow. Deep packet inspection, which underpinned a generation of detectors, no longer has a payload to inspect. Man-in-the-middle decryption is not a practical substitute. It adds latency, removes the privacy guarantee that motivated encryption, and scales poorly at enterprise volumes.

Detection has therefore moved to the metadata that encryption leaves visible. Packet and byte counts, inter-arrival timing statistics, flow duration and connection-level context remain readable from packet headers and switch flow tables. These quantities carry enough behavioural signal to separate normal traffic from many kinds of attack. Detectors over this feature space have progressed from tree ensembles and support-vector classifiers to convolutional, recurrent, Transformer and graph architectures (Chinnasamy et al. 2025; Bilot et al. 2023; Zhong et al. 2024), with strong reported results on UNSW-NB15 (Moustafa and Slay 2015, 2016) and CIC-IDS2017 (Sharafaldin et al. 2018).

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One representational choice is common to almost all of them. Network state is treated as a discrete object. A flow becomes a fixed-length vector drawn from a time window, a node embedding drawn from a periodic graph snapshot, or a token in a packet sequence. The granularity of that discretisation is a hyperparameter fixed before any learning takes place, and it propagates into deployment. Boundary effects appear at window edges. The model holds no representation of network state between sampling instants. Transfer to a monitor that samples at a different rate becomes unreliable. Graph detectors (Lo et al. 2022; Deng and Huang 2025; Lin et al. 2025; Pei et al. 2025) encode which endpoints communicate, but they rebuild their snapshots at fixed intervals and inherit the same constraint.

Attack families absent from training compound the difficulty. Supervised classifiers learn a boundary between labelled attacks and benign traffic and generalise poorly beyond the families their labels cover (Alansary et al. 2025; Ji et al. 2024). Detectors fitted on benign traffic alone are better placed in principle, because they characterise normality rather than any particular attack. Their sensitivity depends on how tightly the learned representation bounds benign behaviour. A representation described at coarse granularity leaves room inside its decision region for traffic whose metadata differs only slightly from benign traffic.

These two limitations have been addressed separately. Removing the dependence on attack labels is the concern of the one-class literature. Carrying topological context is the concern of the graph literature. The discretisation of time is rarely questioned at all. No existing representation combines benign-only learning, graph-topological context and a continuous temporal coordinate over metadata that survives encryption. That combination is the gap addressed here.

Implicit neural representations offer a way to close it (Sitzmann et al. 2020; Essakine et al. 2025). Rather than storing a signal on a grid, an implicit representation learns a function from coordinates to signal values, which can be queried anywhere in the coordinate space. SIREN (Sitzmann et al. 2020) replaces the ReLU activation with a sinusoid. This avoids the spectral bias (Rahaman et al. 2019) that leads conventional networks to fit low-frequency structure preferentially, and it reconstructs images, audio and three-dimensional scenes at high fidelity. ST-INR (Nie et al. 2024) carried the paradigm into road-traffic state, generalising across sensor locations and temporal resolutions.

The obstacle to using implicit representations for network traffic is geometric. Images and road networks are observed over Euclidean coordinates. Traffic is observed over a communication graph that admits no natural Euclidean embedding, and (Essakine et al. 2025) identify this non-Euclidean extension as an open problem for implicit representations generally. A usable coordinate must encode graph position in a differentiable, finite-dimensional form while handling the sign ambiguity, permutation sensitivity and size dependence of spectral descriptors.

This paper introduces GRAPHINR- IDS. It represents the benign behaviour of encrypted traffic as a continuous field over communication topology, capture time and flow metadata. Spectral coordinates derived from the graph Laplacian encode where in the network a flow occurs, and multi-frequency sinusoidal coordinates encode when. A SIREN network reconstructs the flow's metadata from that coordinate together with the observation itself, and a flow is scored by how far its observed metadata departs from the reconstruction. No attack participates in fitting, in residual estimation or in threshold calibration, so every attack family met at test time is unseen by construction. Each flow carries its own coordinate, so no window length is chosen and no verdict depends on which other flows surround it.

Research questions

RQ1 Can continuous graph-temporal reconstruction provide competitive benign-only intrusion detection relative to established one-class methods?
RQ2 Does a pointwise continuous representation reduce dependence on a predefined temporal window and on stream-sampling density?
RQ3 Which architectural components materially contribute to detection performance?
RQ4 How stable are benign residual statistics and calibrated thresholds under chronological distribution shift?
RQ5 How does the method behave when restricted to the features observable under different encrypted-traffic conditions, and when transferred to a second network?

Contributions

1. **Intrusion detection as continuous-field reconstruction.** Detection is formulated as reconstruction of a continuous field over graph topology, capture time and flow metadata, fitted on benign traffic alone. To the best of our knowledge, implicit neural representations have not previously been applied to network intrusion detection.
2. **A spectral graph coordinate for implicit representations.** A Laplacian-eigenvector encoding with sign disambiguation and an explicit slot for endpoints outside the fitted graph maps IP-level topology into a differentiable, finite-dimensional coordinate. Removing this coordinate costs 1.15 percentage points of F1. Retaining it proves costly when endpoints churn, and both effects are quantified.
3. **An evaluation that tests the continuity claim.** The sensitivity of a windowed detector to its window length and to test-time sampling rate is measured alongside that of the continuous field. Eleven one-class detectors, three

classical and eight deep, are compared under a single protocol on two benchmarks, using chronological as well as random splits, multiple seeds and paired significance tests. The chronological protocols expose a calibration failure that the random split conceals.

2 Related work

Research on metadata-based intrusion detection has addressed supervision, temporal modelling and communication topology largely as separate problems. Deep and Transformer detectors capture complex temporal patterns but usually rely on labelled attacks and fixed-length sequences. Graph-based detectors represent endpoint relationships but commonly operate on discrete snapshots. One-class methods remove the dependence on attack labels, although most treat flows as isolated vectors or as members of fixed windows. Implicit neural representations provide a continuous alternative, but their use over communication topology remains unresolved.

2.1 Metadata-Based and sequence models

The dominant line of work treats a flow record, or a short sequence of records, as the unit of classification. Convolutional and recurrent hybrids extract local feature structure and then model longer-range temporal dependence (Sajid et al. 2024; Thaljaoui 2025). Reconstruction autoencoders paired with recurrent components target false-positive suppression (Narmadha and Balaji 2025). Transformer variants adapt language-model machinery to flow records by tokenising either the fields of a record (Ali et al. 2024; Yang and Peng 2025) or its inter-arrival intervals and byte counts (Manocchio et al. 2024). Further designs attend over cloud traffic (Long et al. 2024) or combine convolutional and recurrent stages (Zhang et al. 2025). The review by Chinnasamy et al. (2025) identifies temporal modelling and class imbalance as the persistent difficulties across this literature.

Two assumptions run through the group. The first is supervision. These detectors learn a decision boundary between labelled attack and benign traffic, which serves the families represented in the labels and transfers poorly beyond them. The second is discretisation. Traffic is segmented into windows or sequences before modelling, so the segmentation granularity is fixed in advance and cannot be revised. Detecting an unlabelled family requires abandoning the first assumption. Avoiding a granularity chosen before learning requires abandoning the second.

2.2 Graph-Based intrusion detection

Traffic is naturally graph structured, with IP endpoints as nodes and flows as directed edges. This structure carries information that a flow record alone does not, since identical metadata exchanged between a file server and its clients means something different from the same metadata exchanged between two edge devices. Detectors in this family build on spectral or neighbourhood-aggregation convolutions (Kipf and Welling 2017; Hamilton et al. 2017). E-GraphSAGE (Lo et al. 2022) made edge features first-class citizens of the representation, and EMA-IDS (Deng and Huang 2025) mitigates over-smoothing through multi-hop attention. Later designs add global attention for IoT deployments (Lin et al. 2025), federated training (Jianping et al. 2024) or differential-privacy guarantees (Pei et al. 2025; Wang et al. 2025b; Zhang et al. 2025). Zhong et al. (2024) survey the area and identify over-smoothing in deep graph networks and dynamic topology change as its recurring obstacles.

Most of these detectors remain supervised, and all depend on snapshot graphs rebuilt at fixed intervals. Between consecutive snapshots the model holds no representation of network state. The snapshot period therefore plays the same role for topology that the window length plays for time. A representation in which a flow's graph position is available pointwise, without reference to a snapshot schedule, has not been established.

2.3 Benign-Only anomaly detection

Fitting on benign data alone is the standard response to the open-set problem. The modern methods in this family differ chiefly in what their score measures. Deep SVDD (Ruff et al. 2018) contracts benign representations onto a fixed centre and scores by distance from it. DAGMM (Zong et al. 2018) scores by sample energy under a Gaussian mixture fitted jointly over the latent code and reconstruction-error features. MemAE (Gong et al. 2019) addresses the central weakness of reconstruction-based detection, which is that a sufficiently expressive autoencoder reconstructs anomalies as faithfully as normal data. It does so by forcing the decoder to work from a sparse addressing of learned benign prototypes. USAD (Audibert et al. 2020) trains two decoders adversarially so that one amplifies the error of inputs lying off the benign manifold. CATCH (Wu et al. 2025) separates low-frequency trend from high-frequency variation through channel-aware frequency-domain patching.

Common to this family is a representation of the flow as an isolated vector or as a member of a fixed window. Topology is absent, and time is either discretised or ignored. Removing the dependence on attack labels has been addressed thoroughly. Relaxing the representational assumption alongside it has not.

2.4 Encrypted-Traffic detection

Work aimed specifically at encrypted traffic adds the constraint that no payload is available. Ji et al. (2024) find self-

supervised and reconstruction-based methods to generalise more reliably than supervised classifiers under distribution shift, and (Alansary et al. 2025) identify generalisation to unseen families as the recurring open problem.

ET-SSL (Sattar et al. 2025) is the closest neighbour of the present work. It applies self-supervised contrastive learning to encrypted-traffic anomaly detection without payload access and without labelled attacks. Its score is a relative distance in a learned embedding, so its operating point is a position in that space. A reconstruction residual measured against benign statistics instead yields an operating point that is a quantile of a benign score distribution, and that residual decomposes into a per-feature attribution. Other work in this area remains supervised. CMTSNN (Zhu et al. 2023) classifies encrypted IoT traffic with a BiLSTM–1D-CNN over fixed-length windows, Zang et al. (2024) treat flow features as tokenised sequences, and DIFT (Wang et al. 2025a) employs diffusion for minority-class synthesis rather than for anomaly scoring.

Behdadnia et al. provide the most instructive comparison for the temporal question. In an IT–OT power-grid setting (Behdadnia et al. 2025) they model the network as a traffic-dispersion graph and decompose per-node throughput into wavelet sub-bands. In an IoT setting (Behdadnia et al. 2026) they take packet inter-arrival times as the sole observable and augment their traces with cryptographic processing overhead, reproducing the timing uncertainty that encryption introduces. Their decomposition applies a fixed analytic filter bank to a regularly sampled signal, and their models are trained discriminatively against labelled classes. This permits attribution of an attack vector but requires that the vector has been observed during training. Their treatment of encryption is more direct than either benchmark used here permits.

2.5 Implicit neural representations

Implicit neural representations (Sitzmann et al. 2020) learn a continuous function from coordinates to signal values rather than storing samples on a grid. The SIREN architecture replaces the ReLU activation with a sinusoid, which avoids the spectral bias (Rahaman et al. 2019) of conventional multilayer perceptrons and supports accurate reconstruction of rapidly varying signals. Essakine et al. (2025) survey the field and single out extension to non-Euclidean domains as a key open challenge.

ST-INR (Nie et al. 2024) is the most adjacent application. It maps geographic coordinate and timestamp pairs to road-traffic states and demonstrates low-rank and smoothness properties that benefit imputation and super-resolution. Its coordinate space is Euclidean, which is what a communication graph does not provide. Laplacian eigenvectors supply a finite-dimensional alternative, but they are sign-ambiguous, permutation-sensitive and dependent on graph size, so a usable coordinate requires each property to be handled explicitly. CESNET-TimeSeries24 (Koumar et al. 2025) provides forty weeks of encrypted traffic from 275,000 active IP addresses in an operational ISP, and is the natural target for an evaluation with evolving topology.

2.6 Research gap and comparative positioning

Table 1 sets representative methods against GraphINR-IDS on design properties rather than reported performance, since the cited methods were evaluated by their authors on different datasets and protocols.

No prior method occupies the intersection of the four properties. Supervised graph detectors use topology and non-Euclidean coordinates but require labelled attacks. Self-supervised and reconstruction-based detectors achieve attack-free fitting but discard topology and discretise time. ST-INR supplies a continuous-coordinate paradigm in a Euclidean spatial domain and without a security application. The gap is therefore the absence of a single representation that supports benign-only learning, graph-topological context, a continuous temporal coordinate and metadata-only observation at the same time.

GraphINR-IDS is constructed to occupy that intersection. Three properties of an implicit representation bear on the gap. It is pointwise, so no window length is chosen and a verdict does not depend on how densely the surrounding stream is sampled. Sinusoidal activations avoid the spectral bias of ReLU networks, which matters because benign traffic contains both slow and fast components. The input need only be finite-dimensional, which spectral graph encodings supply without imposing a Euclidean embedding on the topology.

3 Problem formulation

This section states the detection setting and threat model, defines the field the detector approximates, gives the objective it is fitted with and the decision rule applied to its output, and sets out the assumptions each of these carries.

3.1 Detection setting and threat model

The detector observes traffic encrypted under TLS 1.3 or QUIC and therefore sees no payload. What remains visible is flow-level metadata. This comprises packet and byte counts, inter-arrival timing statistics, flow duration, IP header fields and connection-level context. These quantities can be read at line rate from packet headers and switch flow tables, are carried in standard telemetry formats such as NetFlow and IPFIX, and expose no user content. Whether every feature

Table 1 Design comparison of representative related methods and GraphINR-IDS. *Topology*: communication structure used as model input; ✓ marks a non-Euclidean coordinate space, and *Euclidean* marks spatial structure in a Euclidean one. *Continuous t*: time enters as a continuous coordinate rather than through a fixed window or snapshot interval. *Benign-only*: no attack label used at any stage. *Unseen families*: designed to detect attack types absent from training. ✓ = yes; × = no; ∼ = partial. Datasets are those used by the cited authors, not a common protocol

Method	Eval. dataset (as cited)	Model input	Topology	Cont. t	Benign-only	Unseen fam.
Ali et al. (2024)	UNSW-NB15	Tokenised flow records	×	×	×	×
Yang and Peng (2025)	UNSW-NB15	Flow records as text	×	×	×	×
Manocchio et al. (2024)	Multi-NIDS	Tokenised IAT and byte counts	×	×	×	∼
Lo et al. (2022)	UNSW-NB15, ToN-IoT, BoT-IoT	Snapshot graph, edge features	✓	×	×	×
Deng and Huang (2025)	UNSW-NB15	Snapshot graph, multi-hop attn.	✓	×	×	×
Pei et al. (2025)	Multi-NIDS	Snapshot graph, private GNN	✓	×	×	×
Zhu et al. (2023)	Encrypted IoT traffic	Flow windows, BiLSTM–1D-CNN	×	×	×	×
Behdadnia et al. (2025)	IT-OT power-grid traces	Wavelet sub-bands of node throughput	✓	×	×	×
Behdadnia et al. (2026)	P2668-MLIDD, Bot-IoT	Wavelet sub-bands of packet IATs	×	×	×	×
Sattar et al. (2025)	CIC-Darknet, UNSW-NB15	Flow metadata, contrastive embed.	×	×	✓	✓
Ruff et al. (2018)	Image / tabular AD	Feature vector	×	×	✓	✓
Zong et al. (2018)	Tabular AD	Feature vector	×	×	✓	✓
Gong et al. (2019)	Image / video AD	Feature vector, memory bank	×	×	✓	✓
Audibert et al. (2020)	Multivariate time series	Sequence windows	×	×	✓	✓
Wu et al. (2025)	Multivariate time series	Frequency patches	×	×	✓	✓
Nie et al. (2024)	Road traffic	Euclidean coord. + timestamp	Euclidean	✓	✓	×
GraphINR-IDS (this work)	UNSW-NB15, CIC-IDS2017	Spectral graph coord., continuous t, flow metadata	✓	✓	✓	✓

used in the evaluation honours this constraint is a question about the benchmark rather than the method, and Section 5 settles it by audit.

The adversary can inject traffic of arbitrary flow-level characteristics and may control one or more endpoints, but cannot decrypt third-party flows. We assume black-box knowledge. The adversary knows that an anomaly-based metadata detector is in operation and may know that it works by reconstruction, but not its parameters, threshold or architecture. The adversary is not confined to a known attack vocabulary, so the detector must identify traffic from families that were never represented during training. This requirement rules out supervised classification and motivates fitting on benign traffic alone.

Three threat classes lie outside the present scope. An evasion-aware adversary who observes the detector's gradient or score and crafts metadata that stays below threshold is not modelled. Nor is poisoning of the training or calibration partitions, nor extraction or inversion of the deployed model. Man-in-the-middle decryption is also excluded, because it contradicts the privacy assumption that motivates the encrypted-traffic setting.

3.2 Network state field

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ denote the IP-level communication graph, with $\mathcal{V} = \{v_0, v_1, \ldots, v_N\}$ the set of endpoints and $\mathcal{E}$ the set of directed edges. An edge (v_i, v_j) exists when at least one flow from v_i to v_j was observed while fitting. The distinguished node v_0 is not a real endpoint. It is a reserved slot to which any endpoint unseen during fitting is mapped. A directed flow on (v_i, v_j) at continuous time t is observed through a d-dimensional metadata vector $\mathbf{m} \in \mathbb{R}^d$, with $d = 36$ in the UNSW-NB15 evaluation.

The central abstraction is the Network State Field. It maps a flow's position in the network, its time of occurrence and its observed metadata to the metadata a benign flow in that situation would be expected to carry,

$$\Phi : \big(v_i,\ v_j,\ t,\ \mathbf{m}\big) \longmapsto \hat{\mathbf{m}} \in \mathbb{R}^d. \tag{1}$$

Three of the four arguments state where and when the flow occurred and between which endpoints. The fourth is the observation itself. The field returns a reconstruction of what was observed, expressed through a representation fitted exclusively to benign traffic. A flow resembling the benign traffic seen during fitting can be expressed accurately in that representation, and a flow unlike it cannot. The gap between $\mathbf{m}$ and $\hat{\mathbf{m}}$ is therefore the detection signal. Section 3.4 explains why the observation must be supplied as an input for that gap to be informative.

3.3 Learning objective and anomaly decision

The field is approximated by a network f_θ fitted on a corpus $\mathcal{D}_{\text{fit}}$ of benign flows by minimising

$$\theta^* = \arg\min_\theta \ \mathcal{J}_{\text{H}}(\theta) + \lambda_{\text{W}}\,\mathcal{R}_{\text{W}}(\theta) + \lambda_{\text{C}}\,\mathcal{R}_{\text{C}}(\theta), \tag{2}$$

which combines a Huber reconstruction loss $\mathcal{J}_{\text{H}}$, a weight-magnitude penalty $\mathcal{R}_{\text{W}}$ and a residual-compactness penalty $\mathcal{R}_{\text{C}}$. Their forms are given in Section 4.3. No labelled attack enters (2), which is what makes every attack family unseen at test time.

Detection proceeds in three steps. For a test flow, the per-feature squared residual is

$$e_k = (m_k - \hat{m}_k)^2, \qquad k = 1, \ldots, d, \tag{3}$$

with $\hat{\mathbf{m}} = f_{\theta^*}(v_i, v_j, t, \mathbf{m})$. Raw residuals are not comparable across features. A feature the model reconstructs precisely on benign traffic and an intrinsically noisy feature produce errors of quite different magnitude. Each residual is therefore standardised against the mean μ_k and standard deviation σ_k of that feature's residuals on a held-out set of benign validation flows,

$$z_k = \frac{e_k - \mu_k}{\max(\sigma_k,\ \varepsilon)}, \qquad \varepsilon = 10^{-6}, \tag{4}$$

where ε guards against a feature whose residual variance underflows. The flow score is the mean of the κ largest standardised residuals,

$$A = \frac{1}{\kappa} \sum_{k \in \mathcal{T}_\kappa(\mathbf{z})} z_k, \tag{5}$$

with $\mathcal{T}_\kappa(\mathbf{z})$ the index set of the κ largest entries of $\mathbf{z}$. Taking the top κ rather than the mean over all d features concentrates the score on the dimensions where the flow departs from benign behaviour. It also prevents a strong deviation in a few features from being diluted by the many features an attack leaves untouched. We fix $\kappa = \lceil d/4 \rceil = 9$ in advance and select it on no test quantity.

The operating point is set on a calibration partition $\mathcal{D}_{\text{cal}}$ of benign flows, disjoint from all others, by taking the empirical quantile

$$\tau = Q_{1-\alpha}\big(\{A_i : i \in \mathcal{D}_{\text{cal}}\}\big), \qquad \alpha = 0.05, \tag{6}$$

and flagging a flow when $A > \tau$. The quantile rule sets a nominal false-positive rate of α on the benign calibration distribution. It does not fix the realised rate in deployment.

If benign traffic drifts away from the calibration sample, the realised rate departs from α, and Section 6 quantifies how far.

A slow-rate attack may produce residuals that are individually below τ but persistently elevated. For that regime the per-flow score can optionally be pooled over rolling windows per source endpoint,

$$A_{\mathrm{agg}}(t) = \max\big(A(t),\ \bar{A}_{1\,\mathrm{s}}(t),\ \bar{A}_{10\,\mathrm{s}}(t),\ \bar{A}_{60\,\mathrm{s}}(t)\big), \tag{7}$$

where $\bar{A}_w(t)$ averages the per-flow scores from the same source in the interval $(t - w,\ t]$. Pooling makes a verdict depend on neighbouring flows and so forfeits the pointwise property of (5). It is disabled throughout unless stated, and all headline results use the per-flow score.

3.4 Observation-Conditioned reconstruction

Supplying $\mathbf{m}$ as an argument of (1) is what makes the residual informative. It also creates the principal risk to the construction.

The reason for conditioning is that benign endpoint pairs are multi-modal. A field mapping only the graph-temporal coordinate to a flow state would emit a single expected flow for each coordinate, and would then report every legitimate mode other than that one as anomalous. Conditioning on the observation converts the task into constrained reconstruction. The network must express the flow it has been shown, using a representation whose capacity and smoothness are limited by (2), and a flow unlike the benign traffic that shaped the representation cannot be expressed accurately.

The risk is that the network learns the identity map $\hat{\mathbf{m}} = \mathbf{m}$, which reconstructs benign and attack traffic equally well and detects nothing. Neither data-dependent term of (2) prevents this. The Huber term is minimised by the identity map, since that map drives every residual to zero. The residual-compactness term is likewise minimised by it, because that term penalises the variance of the per-sample reconstruction error, and an identity map makes every per-sample error zero. Only the weight-magnitude penalty $\mathcal{R}_{\mathrm{W}}$ acts against identity behaviour, and it does so indirectly. It bounds the gain the network can apply along any input direction, but does not exclude the identity map from the hypothesis class. We therefore make no formal claim that the objective prevents identity collapse. Whether the fitted model behaves as an identity map is an empirical question, and Section 6.3.1 measures it directly.

3.5 Scope and assumptions

The detector operates on flow-level metadata alone, with payload content assumed inaccessible. The graph is built from benign fitting flows, so the topological representation reflects benign communication structure rather than a prior network map. Endpoints absent from it are mapped to the reserved node v_0, and such flows are scored rather than discarded. Threshold calibration through (6) requires a short burn-in of known-benign traffic, which is a standard operational assumption in anomaly detection. Each of these assumptions is a potential failure mode as well as a simplification, and the evaluation is organised to expose them.

4 The GraphINR-IDS model

4.1 Model overview

For each flow, GraphINR-IDS forms a 164-dimensional representation by concatenating a 96-dimensional spectral encoding of the source–destination pair, a 32-dimensional encoding of capture time, and the 36 normalised flow features. The reconstruction network maps this joint representation to an estimate of the benign metadata,

$$\hat{\mathbf{m}} = f_\theta\big[\,\mathbf{PE}(v_i, v_j) \parallel \mathbf{TE}(t) \parallel \tilde{\mathbf{m}}\,\big], \qquad \hat{\mathbf{m}} \in \mathbb{R}^{36}, \tag{8}$$

where $\parallel$ denotes concatenation and $\tilde{\mathbf{m}}$ the min–max normalised metadata vector. The two encoders implement the conceptual arguments of the Network State Field numerically. **PE** turns a pair of endpoints into a position in the communication graph, and **TE** turns a timestamp into a differentiable coordinate. Equation 8 contains everything the model uses to score a flow, and no other flow contributes to $\hat{\mathbf{m}}$.

Figure 1 shows the pipeline, and Table 2 collects the symbols used throughout.

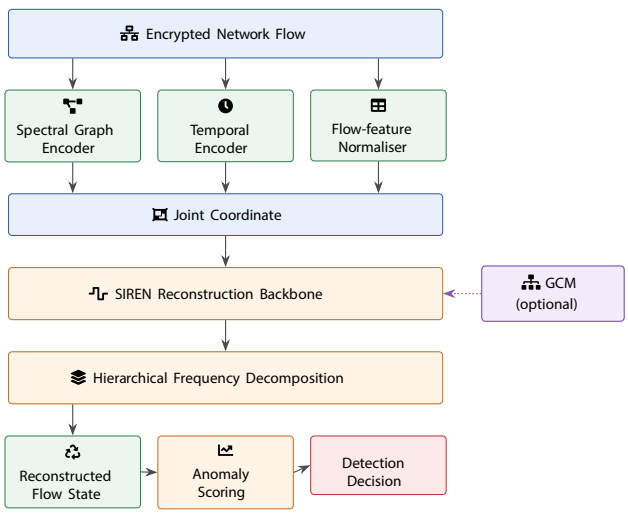


Fig. 1 GraphINR-IDS pipeline: the two coordinate encoders, the SIREN reconstruction network with its optional refinements, and the scoring stage that turns a reconstruction residual into a detection decision

Table 2 Symbols used in Sections 3 and 4

Symbol	Domain	Meaning
Flow, graph and time		
v_i, v_j	$\mathcal{V}$	Source and destination endpoints of a flow
v_0	$\mathcal{V}$	Reserved slot for endpoints unseen while fitting
N, $\lvert\mathcal{E}\rvert$	$\mathbb{N}$	Number of endpoints and of directed edges
t	$\mathbb{R}$	Capture time normalised to the fitting window; exceeds 1 outside it
$\mathbf{m}$, $\tilde{\mathbf{m}}$	$\mathbb{R}^d$	Observed metadata and its min–max normalised form, $d = 36$
$\hat{\mathbf{m}}$	$\mathbb{R}^d$	Reconstructed metadata
Encoders and network		
$\mathbf{L}_G$	$\mathbb{R}^{N\times N}$	Symmetric normalised graph Laplacian
K	$\mathbb{N}$	Laplacian eigenvectors retained, $K = 32$
$\mathbf{p}(v)$	$\mathbb{R}^K$	Spectral embedding of one endpoint
$\mathbf{PE}$	$\mathbb{R}^{3K}$	Spectral encoding of an endpoint pair
F	$\mathbb{N}$	Temporal frequencies, $F = 16$
$\mathbf{TE}$	$\mathbb{R}^{2F}$	Sinusoidal encoding of capture time
Φ	—	Network State Field, the conceptual map
f_θ	—	Its neural approximation, with parameters θ
h, Λ, ω_0	—	SIREN width 256, depth 6, activation scale 30
$\mathbf{g}$	$[0, 1]^d$	Per-feature gate of the two frequency branches
$\mathbf{c}(v)$	$\mathbb{R}^{128}$	Graph context vector used by the modulation module
Scoring and calibration		
e_k	$\mathbb{R}_{\geq 0}$	Squared reconstruction residual of feature k
μ_k, σ_k	$\mathbb{R}$	Residual mean and standard deviation on benign validation flows
z_k	$\mathbb{R}$	Standardised per-feature residual
κ	$\mathbb{N}$	Aggregation width, fixed *a priori* at $\lceil d/4 \rceil = 9$
A	$\mathbb{R}$	Flow score, the mean of the top-κ standardised residuals
α	$(0, 1)$	Target false-positive rate, $\alpha = 0.05$
τ	$\mathbb{R}$	Threshold, the $(1-\alpha)$ quantile of benign calibration scores
B	$\mathbb{N}$	Burn-in size for label-free recalibration

4.2 Graph and temporal coordinates

Spectral graph coordinate

The graph encoder turns an endpoint pair into a continuous position. From the graph $\mathcal{G}$ built on benign fitting flows, the symmetric normalised Laplacian is

$$\mathbf{L}_G = \mathbf{I} - \mathbf{D}^{-1/2}\,\mathbf{A}_s\,\mathbf{D}^{-1/2}, \tag{9}$$

with $\mathbf{A}_s = \frac{1}{2}(\mathbf{A} + \mathbf{A}^\top)$ the symmetrised form of the flow-count-weighted directed adjacency $\mathbf{A}$, and $\mathbf{D}$ its diagonal degree matrix. Symmetrisation guarantees real eigenvectors. Flow direction is preserved because source and destination embeddings are concatenated separately rather than merged.

Let $\{\mathbf{u}_1, \ldots, \mathbf{u}_K\}$ be the $K = 32$ eigenvectors of smallest eigenvalue, excluding the trivial constant eigenvector. The embedding of a node is the vector of its entries in those eigenvectors, $\mathbf{p}(v_i) = [u_1(v_i), \ldots, u_K(v_i)]^\top$. Laplacian eigenvectors are determined only up to a sign. The sign of each $\mathbf{u}_k$ is fixed by requiring its largest-magnitude entry to be positive (Lim et al. 2023), which makes the encoding deterministic across runs. The edge encoding combines the two endpoints,

$$\mathbf{PE}(v_i, v_j) = \left[\mathbf{p}(v_i) \,\|\, \mathbf{p}(v_j) \,\|\, \mathbf{p}(v_i) \odot \mathbf{p}(v_j)\right] \in \mathbb{R}^{3K}, \tag{10}$$

with $\odot$ the Hadamard product, giving 96 dimensions at $K = 32$. The product term allows the network to distinguish a pair of endpoints from the two endpoints considered separately.

Endpoints absent from the fitting graph are mapped to the reserved node v_0, which is assigned $\mathbf{p}(v_0) = \mathbf{0}$. The product term then vanishes and half the concatenation is zero, placing the flow at a nominally neutral position rather than attributing it to an arbitrary known host. Section 6.3.2 measures the cost of this choice as endpoint churn increases.

Temporal coordinate

Capture time enters through a sinusoidal encoding. Let t^{raw} be the flow start time in seconds. Times are normalised against the fitting window,

$$t = \frac{t^{\text{raw}} - t^{\text{raw}}_{\min}}{t^{\text{raw}}_{\max} - t^{\text{raw}}_{\min}}, \tag{11}$$

using the earliest and latest timestamps of the fitting partition. The fitting window therefore maps to [0, 1], and the constants are estimated on benign data like every other quantity in the pipeline. Normalisation is necessary because the raw epoch, of order 1.4×10^9 s, would alias all but the smallest frequencies. The encoding uses $F = 16$ log-linearly spaced frequencies $\{f_1, \ldots, f_F\}$ between 1 Hz and 10 kHz,

$$\mathbf{TE}(t) = \big[\sin(2\pi f_1 t), \ldots, \sin(2\pi f_F t), \cos(2\pi f_1 t), \ldots, \cos(2\pi f_F t)\big] \in \mathbb{R}^{2F}, \tag{12}$$

which resolves structure from whole-window trends down to intervals of order 10^{-4} of the fitting span.

Equation 11 is not clipped, so a flow captured after the fitting window yields $t > 1$. The encoding remains bounded because it is periodic, but it does not extrapolate. For $t > 1$ it returns phases that also occur within [0, 1], so the field generalises by aliasing onto phases it has already seen rather than by extending a trend. This is a property of sinusoidal positional encodings generally, and Section 6.4.1 measures its consequences for deployment.

4.3 SIREN reconstruction network

The reconstruction network is a SIREN (Sitzmann et al. 2020), a multilayer perceptron with sinusoidal activations. It maps the 164-dimensional coordinate $\mathbf{x}$ of (8) to $\hat{\mathbf{m}}$ through

$$\mathbf{h}_0 = \sin\big(\omega_0(\mathbf{W}_0\mathbf{x} + \mathbf{b}_0)\big), \tag{13}$$

$$\mathbf{h}_\ell = \sin\big(\omega_0(\mathbf{W}_\ell\mathbf{h}_{\ell-1} + \mathbf{b}_\ell)\big), \quad \ell = 1, \ldots, \Lambda - 1, \tag{14}$$

$$\hat{\mathbf{m}} = \mathbf{W}_\Lambda\mathbf{h}_{\Lambda-1} + \mathbf{b}_\Lambda, \tag{15}$$

with $\Lambda = 6$ hidden layers of width $h = 256$ and frequency parameter $\omega_0 = 30$. Weights follow the SIREN initialisation, $\mathbf{W}_0 \sim \mathcal{U}(-1/n,\ 1/n)$ and $\mathbf{W}_\ell \sim \mathcal{U}(-\sqrt{6/n}/\omega_0,\ \sqrt{6/n}/\omega_0)$ for $\ell \geq 1$, with n the layer's input width. This initialisation keeps forward and gradient norms stable (Sitzmann et al. 2020).

Sinusoidal activations are chosen for a representational reason. ReLU networks exhibit a spectral bias (Rahaman et al. 2019) towards low-frequency structure, and a detector whose sensitivity depends on how tightly it bounds benign behaviour should not be biased against the rapidly varying components of that behaviour.

The network is fitted by (2). Its three terms take the following forms. The reconstruction term is a Huber loss (Huber 1964) with $\delta = 1$,

$$\mathcal{J}_{\text{H}}(\theta) = \frac{1}{|\mathcal{B}|}\sum_{i\in\mathcal{B}} \mathcal{L}_\delta\big(\mathbf{m}_i, \hat{\mathbf{m}}_i\big), \tag{16}$$

which is less sensitive to large per-feature errors than mean squared error and stabilises fitting under the feature-level heterogeneity of flow records. The second term penalises weight magnitude,

$$\mathcal{R}_{\text{W}}(\theta) = \frac{\sum_\ell \|\mathbf{W}_\ell\|_F^2}{\sum_\ell \operatorname{card}(\mathbf{W}_\ell)}, \qquad \lambda_{\text{W}} = 10^{-4}, \tag{17}$$

summing over weight matrices of rank at least two. Normalising by parameter count keeps the penalty comparable across ablations of different size. Bounding weight magnitude bounds the gain the network can apply along any input direction, and is a weaker constraint than a spectral-norm bound on the Lipschitz constant. The third term penalises the batch variance of the per-sample reconstruction error,

$$\mathcal{R}_{\text{C}}(\theta) = \operatorname{Var}_{i\in\mathcal{B}}\left[\frac{1}{d}\sum_{k=1}^{d}(m_{ik} - \hat{m}_{ik})^2\right], \qquad \lambda_{\text{C}} = 0.1, \tag{18}$$

which concentrates the benign residual distribution. A tight benign residual distribution makes the quantile threshold of (6) informative, since the spread of benign scores rather than their mean determines how far an attack must deviate before crossing τ.

4.4 Architectural refinements

Two optional modules sit on top of the reconstruction network. Each is a hypothesis about what the base representation may be missing, and each is evaluated by removal in Section 6.3.

Hierarchical frequency decomposition

Flow features may vary at different temporal scales, with session durations changing slowly and inter-packet timing rapidly. A single ω_0 commits the whole reconstruction to one spectral range. This module tests whether separate frequency branches improve reconstruction. Two SIREN branches of the form of (13)–(15) share the input coordinate but not their weights, one at $\omega_L = 10$ and one at $\omega_H = 60$, bracketing the reference $\omega_0 = 30$ at a third and twice its value. A gate computed from the coordinate by a three-layer perceptron of width 64,

$$\mathbf{g} = \sigma\big(\mathbf{W}_3\,\text{ReLU}(\mathbf{W}_2\,\text{ReLU}(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2) + \mathbf{b}_3\big) \in [0, 1]^d, \tag{19}$$

combines them per feature,

$$\hat{\mathbf{m}} = \mathbf{g} \odot \mathbf{y}_L + (\mathbf{1} - \mathbf{g}) \odot \mathbf{y}_H, \tag{20}$$

where $\mathbf{y}_L$ and $\mathbf{y}_H$ are the two branch outputs. The gate operates per feature because a byte count and an inter-packet interval need not share a frequency branch. It is conditioned on the input coordinate rather than on the branch outputs. Conditioning on the outputs would allow the gate to select whichever branch reconstructs an attack flow more accurately, which is behaviour a reconstruction-based detector must not have. No frequency search was performed, and the pair (ω_L, ω_H) remains a hyperparameter that may repay per-deployment tuning.

Graph-Conditioned modulation

The spectral coordinate places a flow in the graph but says nothing about the current behaviour of its neighbours. This module tests whether local graph context carries information beyond that coordinate. A two-layer graph attention network (Veličković et al. 2018) of width 128 runs over the precomputed spectral embeddings and produces a context vector $\mathbf{c}(v)$ per node. For a flow on (v_i, v_j), a generator maps $[\mathbf{c}(v_i) \,\|\, \mathbf{c}(v_j)]$ to a scale and shift applied after each sinusoidal layer,

$$\mathbf{h}'_\ell = \boldsymbol{\gamma}_\ell \odot \mathbf{h}_\ell + \boldsymbol{\beta}_\ell, \qquad \boldsymbol{\gamma}_\ell = \mathbf{1} + \eta \tanh(\cdot), \quad \boldsymbol{\beta}_\ell = \eta \tanh(\cdot), \tag{21}$$

with $\eta = 0.1$. This follows feature-wise linear modulation (Perez et al. 2018), with one difference that matters inside a SIREN. The modulation is bounded, so conditioning perturbs the activations by at most 10 %. Unbounded modulation interacts badly with the ω_0 scaling, because a multiplicative factor on a pre-activation is equivalent to changing that layer's effective frequency.

4.5 Scoring and calibration

Deployment separates into a one-off fitting stage and a per-flow inference stage, set out in Algorithm 1. Fitting builds the graph, extracts and caches the spectrum, fits the normalisers and the network, then estimates the residual statistics on benign validation flows and the threshold on a disjoint benign calibration partition. Every quantity is derived from benign data, and the four partitions are mutually disjoint.

Inference reduces to a table lookup, one forward pass, and an order statistic over 36 numbers. Two consequences follow. A verdict depends on the flow alone and is unaffected by how the surrounding stream is batched or sampled, provided the optional pooling of (7) is left disabled. The threshold is a benign quantile, so the attack mix in the test partition does not influence where the operating point sits, although drift in the benign traffic does.

Algorithm 1 Fitting and inference. All fitted quantities derive from benign flows.

Fitting ▷ inputs $\mathcal{D}_{\text{fit}}, \mathcal{D}_{\text{val}}, \mathcal{D}_{\text{cal}}$, all benign
1: build $\mathcal{G}$ from the endpoint pairs of $\mathcal{D}_{\text{fit}}$; form $\mathbf{L}_G$ and cache its K smallest eigenpairs with signs fixed
2: fit the feature and time normalisers on $\mathcal{D}_{\text{fit}}$; assign v_0 a zero spectral row
3: $\theta^* \leftarrow \arg\min_\theta$ (2) over $\mathcal{D}_{\text{fit}}$
4: $(\mu_k, \sigma_k) \leftarrow$ mean and standard deviation of e_k over $\mathcal{D}_{\text{val}}$ ▷ (3)
5: $\tau \leftarrow Q_{1-\alpha}$ of the scores of $\mathcal{D}_{\text{cal}}$ ▷ (6)

Inference ▷ one flow $(v_i, v_j, t, \mathbf{m})$
6: $\mathbf{x} \leftarrow [\mathbf{PE}(v_i, v_j) \,\|\, \mathbf{TE}(t) \,\|\, \tilde{\mathbf{m}}]$ ▷ v_0 if an endpoint is unseen
7: $e_k \leftarrow (m_k - \hat{m}_k)^2$ with $\hat{\mathbf{m}} = f_{\theta^*}(\mathbf{x})$
8: $z_k \leftarrow (e_k - \mu_k)/\max(\sigma_k, \varepsilon)$; $A \leftarrow$ mean of the κ largest z_k
9: **return attack** if $A > \tau$, otherwise **benign**

4.6 Computational characteristics

Cost separates along the same fitting and inference line. Graph construction is linear in the number of fitting flows. The spectral stage extracts the K smallest eigenpairs of a sparse symmetric Laplacian by Lanczos iteration at $O(K|\mathcal{E}|)$ per iteration with $O(KN)$ memory, and is computed once and cached.

Inference is independent of the number of endpoints N. The graph encoder is a table lookup, the temporal encoder costs $2F$ transcendental evaluations, and a forward pass is $O(d_{\text{in}}h + \Lambda h^2 + hd)$, or about 0.4 M multiply–accumulate operations per branch. Resident memory is the network plus the $K \times N$ eigenvector table, which occupies $4KN$ bytes in single precision, or 128 MB at $N = 10^6$ and $K = 32$. Scaling to networks of 10^5 to 10^6 endpoints is therefore governed by the eigendecomposition and by how often topology change forces it to be recomputed, not by the per-flow path.

One consequence bounds what this evaluation can establish about scale. The graph used here has 48 endpoints and 218 directed edges, so with $K = 32$ the spectral encoding is close to a complete eigenbasis. The topological coordinate is nearly lossless at this scale and correspondingly uninformative about the regime $K \ll N$.

5 Experimental methodology

This section sets out what the evaluation measures and under what conditions. It covers which features a monitor could observe under encryption, the three protocols and their benchmarks, the model and baseline configurations, and the statistical treatment applied to the results.

5.1 Feature observability under encryption

A detector claiming to work on encrypted traffic must be built from quantities that encryption leaves visible. The bench-

mark used here does not guarantee that. UNSW-NB15 was captured without transport encryption, so every field in its feature vector is populated regardless of whether a deployed monitor could compute it. We therefore audited all 36 features against the observation each one requires. Table 3 summarises the result.

The audit divides the features into three tiers. Twenty-four follow from the outer IP header and packet arrival times alone, and survive TLS 1.3, QUIC, DTLS, IPsec and VPN encapsulation. Seven require the TCP header in the clear, which TLS over TCP provides but QUIC, DTLS over UDP and tunnels do not. Five describe HTTP and FTP session structure, require plaintext application data, and are unavailable under every protocol this paper targets. Twelve of the 36 features are therefore unavailable in the setting the method addresses. Detection is consequently reported for all three nested sets rather than for the benchmark set alone.

Observability is not the only way encryption affects these features. The surviving features are also perturbed, and not uniformly across protocols. Record padding and the cipher's block structure quantise packet sizes, which compresses the variance of the mean-size features. Handshake cost shifts the early part of a flow asymmetrically. A TLS 1.3 full handshake adds a round trip before application data, while QUIC 0-RTT resumption removes one, so duration and the first inter-packet intervals differ between protocols even for identical application behaviour. Cryptographic processing adds a small load-dependent delay that inflates the jitter features. Behdadnia et al. (2026) model this directly and find inter-arrival statistics still discriminative once it is accounted for. Tunnelling has the largest effect, because it changes what an address denotes. Inside an IPsec or VPN tunnel the visible endpoints are gateways, so both the connection-context counters and the communication graph describe the tunnel rather than the hosts, and GraphINR-IDS then applies at tunnel-endpoint level. No trace available to us isolates these effects individually, so Section 6.5.2 studies them by controlled perturbation of the same feature groups.

5.2 Datasets and evaluation protocols

Three protocols of increasing difficulty are used across two benchmarks. A random split of a single benchmark cannot separate a detector that has learned benign behaviour from one that has learned the benchmark. The later protocols therefore withhold first temporal and then network similarity.

Protocol I is a random benign split of UNSW-NB15 (Moustafa and Slay 2015), which labels flows across nine attack families. We use the four raw CSV captures rather than the pre-split distribution, because only the raw files retain IP addresses and the `Stime` timestamp. The first 150,000 records of each capture give 600,000 flows, of which 509,393 are benign. From these, 80,000 benign and 25,000 attack flows are drawn and partitioned as in Table 4. Five seeds redraw the partition and reinitialise the model.

Protocol II splits the same corpus chronologically. UNSW-NB15 was captured in two campaigns separated by 630 hours. This protocol fits on the first and tests on the second, so every test flow post-dates every fitting flow by roughly 26 days. It is a closer approximation to temporal deployment shift than a random split provides, since an operator calibrates once and the detector then runs against a later period. It also exercises the temporal encoding outside its fitted interval.

Protocol III moves to a second network. CIC-IDS2017 (Sharafaldin et al. 2018) provides five capture days, of which Monday alone is attack-free. The detector is fitted on Monday benign traffic and tested on Tuesday to Friday. All fourteen attack classes are unseen at fitting time, the capture infrastructure differs from UNSW-NB15, and the benign traffic shifts across days. One structural limitation constrains this protocol. The publicly redistributed CIC-IDS2017 files carry 78 derived features and the label, but neither IP addresses

Table 3 Observability of the 36 UNSW-NB15 flow features under encryption. Tiers are defined by what an on-path monitor must be able to read; ✓ marks a feature that remains computable and × one that does not. Because UNSW-NB15 was captured without transport encryption, the twelve Tier B and Tier C features are populated in the benchmark although a deployed detector would not observe them, which is why detection is reported separately for each nested set. Definitions follow the dataset documentation (Moustafa and Slay 2015)

Tier	#	Requires	Representative features	TLS/TCP	QUIC, DTLS	Tunnel	Set
A	24	Outer IP header and packet arrival times only	`sbytes`, `dbytes`, `Spkts`, `Dpkts`, `smeansz`, `Sload`, `dur`, `Sjit`, `Sintpkt`, `sttl`, and the `ct_*` connection-context counters	✓	✓	✓	A
B	7	TCP header in the clear	`tcprtt`, `synack`, `ackdat`, `swin`, `dwin`, `sloss`, `dloss`	✓	×	×	A+B
C	5	Plaintext application payload	`trans_depth`, `res_bdy_len`, `ct_flw_http_mthd`, `is_ftp_login`, `ct_ftp_cmd`	×	×	×	A+B+C

Tier A is the 24-feature set observable under every protocol considered; Tier A+B the 31-feature set observable under TLS over TCP; Tier A+B+C the full 36-feature benchmark set. The complete per-feature list accompanies the code and data

Table 4 Data partitions. Every partition on which a quantity is fitted contains benign flows only, and the four partitions are mutually disjoint. Protocol I redraws the benign partition at every seed. Under Protocols II and III the fitting partitions are taken in capture order from the earlier period and are therefore fixed; only the sampling of the test partition varies with the seed. The fitting partitions are the same size under all three protocols, so the protocols differ in how the data is divided and not in how much the detector is given

Protocol	Partition	Benign	Attack	Fitted quantity
I UNSW-NB15 random	Training	64,000	0	f_θ, feature ranges, graph, spectrum, t-window
	Validation	5,000	0	residual statistics (μ_k, σ_k)
	Calibration	5,000	0	threshold τ
	Test	6,000	25,000	—
II UNSW-NB15 cross-session	Training	64,000	0	as above
	Validation	5,000	0	residual statistics
	Calibration	5,000	0	threshold τ
	Test	20,000	40,000	—
III CIC-IDS2017 by capture day	Training	64,000	0	as above (graph and t inert)
	Validation	5,000	0	residual statistics
	Calibration	5,000	0	threshold τ
	Test	20,000	40,000	—

Protocol I test partitions are attack-heavy by construction, which inflates accuracy relative to an operational base rate; AUPRC and the realised false-positive rate are reported alongside for this reason. For Protocol III, training, validation and calibration are drawn in capture order from Monday, the only attack-free day, and the test partition is drawn from Tuesday to Friday

nor flow timestamps, so the spectral and temporal coordinates cannot be instantiated. Protocol III therefore evaluates the reconstruction detector rather than the full architecture, against the same baselines under identical conditions.

Preprocessing is identical throughout. Features are scaled to $[0, 1]$ by min–max normalisation, with the range taken from the fitting partition and test values outside it clipped. The communication graph is built from benign fitting flows only, adding a directed edge for each unique source–destination pair weighted by flow count. The Laplacian and its 32 smallest eigenpairs are computed once and cached, and are never updated with test information. Every fitted quantity derives from benign data, and the partitions on which they are fitted are mutually disjoint.

5.3 Model configuration and fitting

Table 5 lists the configuration used in all reported experiments. The model is fitted for 50 epochs on benign traffic using AdamW (Loshchilov and Hutter 2019) at learning rate 5×10^{-5}, with $\beta_1 = 0.9$, $\beta_2 = 0.999$ and weight decay 10^{-4}. The learning rate follows a cosine annealing schedule (Loshchilov and Hutter 2017) whose first cycle spans the whole budget, so no warm restart is triggered. Minibatches hold 1,024 flows sampled uniformly, and gradients are clipped to unit norm. All results come from a 16-thread commodity CPU host, on which one 50-epoch fit of the full model takes about seven minutes of wall-clock time with four threads. Spectral eigenvectors are computed deterministically and cached, and the threshold is derived programmatically from the calibration-benign score distribution, with no manual tuning.

5.4 Comparison methods

Eleven one-class detectors are evaluated under conditions identical to those of GraphINR-IDS. All use the same benign fitting partition, the same 36 normalised features, the same benign calibration partition and the same threshold rule, and none sees an attack label at any stage. The deep baselines additionally share the preprocessing, normalisation and early-stopping regime of the proposed model, so differences between them are attributable to the detector rather than to how it was trained. One asymmetry remains. The graph and time coordinates are specific to the proposed model, and the tabular baselines cannot consume them.

Three baselines are classical. **Isolation Forest** uses 200 estimators. **Local Outlier Factor** uses $k = 20$ neighbours in novelty mode, fitted on a 20,000-flow subsample. The **One-Class SVM** uses an RBF kernel with γ from the median heuristic on a 5,000-flow subsample and $\nu = 0.05$.

Eight are deep. The **Dense Autoencoder** is $36 \to 64 \to 32 \to 64 \to 36$ with ReLU activations and MSE loss. The **LSTM Autoencoder** uses a two-layer encoder of width 64 over windows of 16 consecutive flows with a mirrored decoder. Each flow is scored by the mean reconstruction error over all stride-one windows containing it, so its score does not depend on which other flows share a window with it. **Deep SVDD** (Ruff et al. 2018), **DAGMM** (Zong et al. 2018), **MemAE** (Gong et al. 2019) and **USAD** (Audibert et al. 2020) are the modern one-class detectors described in Section 2.3. The **Transformer Autoencoder** treats each feature as a token and reconstructs the vector with a two-layer pre-norm encoder (Vaswani et al. 2017), capturing inter-feature dependence without imposing a window. The **diffusion detector**

Table 5 Model and training configuration. All experiments use this configuration unless a component is explicitly disabled in the ablation or a feature tier is restricted in Section 6.5.1

Parameter	Value	Where defined
Input coordinate		
Flow features $\tilde{\mathbf{m}}$	36 (min–max normalised)	Table 3
Spectral eigenvectors K	32	(10)
Edge encoding dimension	96	(10)
Temporal frequencies F	16 (1 Hz–10 kHz, log-spaced)	(12)
Temporal encoding dimension	32	(12)
Time normalisation	training-window min–max	(11)
Total input dimension	164	(8)
SIREN backbone		
Hidden layers / width	6 / 256	(13)–(15)
Output dimension	36	(15)
Frequency parameter ω_0	30	(14)
Graph-conditioned modulation		
GNN layers / hidden width	2 / 128	(21)
Modulation	bounded affine, $\eta = 0.1$	(21)
Hierarchical frequency decomposition		
Low / high branch ω_L, ω_H	10, 60	(20)
Gate hidden width	64, per-feature output	(19)
Parameter counts		
Full model	1,983,212	—
Without GCM	777,964	—
Without HFD	1,585,700	—
Backbone only	380,452	—
Training		
Optimiser / learning rate	AdamW / 5×10^{-5}	Section 5.3
Weight decay / gradient clip	10^{-4} / 1.0	Section 5.3
Schedule / epochs / batch	cosine anneal / 50 / 1,024	Section 5.3
Reconstruction loss	Huber, $\delta = 1$	(16)
Weight-magnitude penalty λ_W	10^{-4}	(17)
Residual-compactness penalty λ_C	0.1	(18)
Scoring		
Aggregation	top-κ, $\kappa = 9$ (fixed *a priori*)	(5)
Residual standardisation	per-feature mean and SD	(4)
Threshold	95th percentile of calibration-benign scores	(6)
Target FPR α	0.05	(6)

fits a denoising diffusion model to the benign feature distribution and scores by expected noise-prediction error at a fixed step, averaged over four noise draws.

Two families are discussed but not implemented. PatchCore-style memory-bank detection depends on a pretrained feature extractor and on spatial patch structure, neither of which has an analogue in a 36-dimensional flow record. MemAE is the closest memory-based detector that does transfer. DIFT (Wang et al. 2025a) uses diffusion for supervised minority-class synthesis rather than for anomaly scoring, and the diffusion detector above is its one-class counterpart.

5.5 Evaluation metrics and statistical analysis

We report accuracy, precision, recall, F1, AUROC, AUPRC and the realised false-positive rate. AUPRC is included because the test partitions are attack-heavy by construction, and threshold-free ranking quality at a realistic operating point is better summarised by precision–recall than by the ROC curve.

Per-family detection rates are reported for the proposed model, and they carry all the unseen-family information this design admits. No attack family participates in fitting, so

Table 6 Number of independent runs behind each experiment, and why. A run redraws the data partition, reinitialises the model and reorders the minibatches. Where a count is low the reason is compute rather than choice, and the corresponding table suppresses any ranking claim

Experiment	Script	Runs	Reason for that number
Protocol I comparison and ablation	`exp1`	5	Primary result; five fits give a usable spread for paired tests.
Baseline comparison	`exp2`	3	Eleven detectors refitted per seed; three bounds the compute.
Identity diagnostics	`exp5`	3	Diagnostic, not a ranking; three confirms the sign is stable.
Continuity experiments	`exp4`	3	Four sub-experiments per seed over a shared fitted model.
Topology and out-of-vocabulary	`exp6`	3	Each seed redraws the split, changing which edges are unseen.
Perturbation study	`exp7`	3	Seven conditions at four intensities, on a fixed fitted model.
Drift and recalibration	`exp11`	3	Five burn-in sizes and two time-coordinate settings per seed.
Scoring-stage sweep	`exp8`	2	Robustness check on a fixed model; declared three, two completed.
Encryption-tier study	`exp10`	2	Three nested feature sets, each requiring a separate fit.
Protocol II (cross-session)	`exp3`	2	Every one of twelve detectors is refitted per seed.
Protocol III (cross-day)	`exp3`	1	As above, on a second corpus; no dispersion is claimed.
Bootstrap intervals	`exp12`	1	Interval is over test flows, not over fits (Table 8).

every family in the test partition is unseen. The threshold is a benign quantile, so no attack family influences it. The per-family detections partition the global true-positive count, so their aggregate is recall. A leave-one-family-out procedure has no purchase here, because no family is present in fitting that could be withheld, and no separate zero-day rate is therefore reported.

Results are means over independent seeds, with the standard deviation after ± where three or more seeds are available. A seed redraws the partition, reinitialises the model and reorders the minibatches, so the spread reflects sampling and optimisation variability together. Comparisons are paired on the seed. Two sources of uncertainty are distinguished throughout. Seed-to-seed variability is estimated by repeated fitting, and flow-level sampling variability by a stratified bootstrap over the test partition with 2,000 resamples at a fixed seed. Where a difference is smaller than the seed-to-seed spread, no ranking is claimed. Table 6 gives the number of runs behind each experiment and the reason for it, and the count also appears in each results table.

6 Results

All experiments share a common basis. Fitting uses benign flows only, the threshold is the 95th percentile of benign calibration scores, and seeds are paired across methods. Individual experiments aggregate over different numbers of

Table 7 Detection performance on UNSW-NB15 under Protocol I (random benign split). Every detector is fitted on the same benign training partition and thresholded at the 95th percentile of the same benign calibration partition; none sees an attack label. Values are mean ± SD over independent seeds. The best value in each of the first six columns is in **bold**; FPR is not emphasised because it is a calibration target rather than a quantity to maximise, and is reported against a 5.00 % target

Method	Acc.	Prec.	Recall	F1	AUROC	AUPRC	FPR
Transformer Autoencoder	40.94 ± 3.78	95.69 ± 0.89	28.00 ± 4.61	43.20 ± 5.55	87.97 ± 6.81	94.65 ± 1.48	5.14 ± 0.45
USAD	48.29 ± 10.38	96.61 ± 1.37	37.06 ± 12.89	52.72 ± 14.51	82.18 ± 6.42	93.88 ± 2.68	4.92 ± 0.63
Isolation Forest	52.14 ± 4.66	97.18 ± 0.33	41.86 ± 5.83	58.36 ± 5.87	94.33 ± 0.60	96.69 ± 0.26	5.03 ± 0.52
Local Outlier Factor	58.65 ± 15.75	97.46 ± 0.72	49.93 ± 19.64	64.60 ± 16.88	78.47 ± 5.56	94.31 ± 1.54	5.02 ± 0.46
One-Class SVM	71.85 ± 2.00	98.19 ± 0.21	66.32 ± 2.38	79.15 ± 1.77	94.72 ± 0.99	97.36 ± 0.40	5.08 ± 0.41
DAGMM	77.18 ± 14.86	98.49 ± 0.40	72.77 ± 18.46	82.78 ± 13.25	90.89 ± 5.17	97.23 ± 1.24	4.44 ± 0.39
MemAE	88.66 ± 7.06	98.72 ± 0.24	87.06 ± 8.67	92.37 ± 5.14	97.40 ± 1.02	98.94 ± 0.35	4.64 ± 0.38
Deep SVDD	94.53 ± 2.69	**98.79 ± 0.09**	94.37 ± 3.35	96.51 ± 1.78	99.06 ± 0.47	99.73 ± 0.11	4.79 ± 0.38
Dense Autoencoder	94.73 ± 0.99	98.72 ± 0.05	94.69 ± 1.20	96.66 ± 0.65	98.65 ± 0.10	99.57 ± 0.06	5.11 ± 0.15
Diffusion (DDPM) detector	98.38 ± 0.28	98.79 ± 0.04	99.20 ± 0.38	98.99 ± 0.17	98.31 ± 0.29	98.78 ± 0.18	5.05 ± 0.20
LSTM Autoencoder	**98.92 ± 0.08**	98.78 ± 0.10	**99.90 ± 0.01**	**99.34 ± 0.05**	96.66 ± 0.31	97.64 ± 0.34	5.13 ± 0.43
GraphINR-IDS	98.48 ± 0.18	98.78 ± 0.04	99.34 ± 0.23	99.06 ± 0.12	**99.66 ± 0.04**	**99.88 ± 0.02**	5.13 ± 0.18

All values in percent. Baselines over 3 seeds, GraphINR-IDS over 5

seeds, so the full model under Protocol I appears as 99.06 % F1 over five seeds, 99.13 % over three and 99.14 % over two. These are the same configuration at $\kappa = 9$, and every table states its own run count.

6.1 Comparative benign-only detection

Comparative performance was evaluated against eleven one-class detectors fitted and calibrated on identical benign partitions. All twelve methods used the same 64,000 benign fitting flows and the same 95th-percentile threshold over the same 5,000 benign calibration flows. None observed an attack label at any stage. Table 7 reports the outcome.

GraphINR-IDS reaches 99.06 ± 0.12 % F1 at a 5.13 % false-positive rate. It attains the highest AUROC (99.66 ± 0.04 %) and AUPRC (99.88 ± 0.02 %) of the twelve, and the second-highest F1. The LSTM Autoencoder attains the highest F1 at 99.34 %. The two observations are compatible. AUROC and AUPRC measure how well a detector orders flows across all operating points, while F1 measures one point on that ordering. A detector can place slightly more attacks above a particular threshold while ordering the remainder less well. The LSTM Autoencoder's AUROC sits roughly three points below that of the proposed model.

The benchmark itself limits what this comparison establishes. Five of the twelve detectors fall between 95 and 99.5 % F1, and the three strongest span 0.35 percentage points, which is comparable to the spread across seeds within a single method. A stratified bootstrap over the test flows of one fitted model (Table 8) initially suggests otherwise. Resampling the 31,000 test flows gives intervals that do not overlap between the three strongest detectors, so a single fitted instance of each would appear reliably separable. That separability does not survive retraining. The seed-to-seed spread across those three methods is wider than any individual model's flow-level interval, because it captures variation from the data partition and the optimisation as well as from the test sample. Ranking claims in this paper therefore rest on the seed-to-seed comparison, and differences of a few tenths of a percentage point on this benchmark are not treated as meaningful.

Further down the table the gaps are wide enough to interpret, and the classical detectors behave as this feature space predicts. Isolation Forest and Local Outlier Factor recover less than 60 % of attacks at the calibrated threshold. Axis-aligned partitioning and local density estimation both struggle when attack regions abut benign clusters in 36 dimensions. One-Class SVM is stronger but stays below every reconstruction-based method except USAD and the Transformer Autoencoder. Those two are the weakest deep detectors, and specifically in ranking rather than thresholding. USAD's AUROC lies 8.71 to 16.88 points below the other deep methods, and the Transformer Autoencoder's 2.92 to 11.09 points below them. Both were designed for structure this setting lacks, USAD for multivariate time series with temporal windows and the Transformer for sequences long enough for self-attention to exploit. A single 36-dimensional flow record supplies neither. Figure 2 plots the comparison, and Fig. 3 shows the operating point and the score distributions behind it for the proposed model.

Table 9 breaks detection down by attack family under the single globally calibrated threshold. Every family listed is unseen at fitting time, and these rates aggregate to the recall reported above.

6.2 Sensitivity to temporal discretisation

Dependence on temporal discretisation was probed through four manipulations of the temporal coordinate and of the comparator's window, summarised in Table 10 and plotted in Figs. 4 and 5. The outcomes do not uniformly favour the proposed design.

Table 8 Stratified bootstrap 95% confidence intervals under Protocol I, seed 0 (2,000 resamples, drawn separately from the benign and attack test flows at their observed sizes). This reflects sampling variability at the flow level, distinct from the seed-to-seed variability of Table 7, which reflects data-partition and training variability. Both are properties of the same detector: the interval here is conditional on the single fitted model at this seed. Precision, recall and AUPRC intervals are given in the accompanying repository

Method	F1 [95% CI]	AUROC [95% CI]
USAD	35.97 [35.27, 36.63]	75.63 [74.75, 76.50]
Transformer Autoencoder	49.30 [48.64, 49.93]	80.27 [79.64, 80.92]
Isolation Forest	51.72 [51.09, 52.35]	94.43 [93.93, 94.91]
Local Outlier Factor	61.63 [61.04, 62.22]	79.70 [79.16, 80.24]
One-Class SVM	79.71 [79.29, 80.15]	95.19 [94.80, 95.58]
DAGMM	90.52 [90.24, 90.78]	93.13 [92.78, 93.48]
MemAE	95.11 [94.92, 95.31]	97.75 [97.45, 98.03]
Dense Autoencoder	97.40 [97.27, 97.54]	98.72 [98.54, 98.89]
Deep SVDD	98.01 [97.89, 98.13]	99.42 [99.34, 99.50]
Diffusion (DDPM) detector	98.82 [98.73, 98.91]	98.38 [98.09, 98.65]
LSTM Autoencoder	99.38 [99.31, 99.45]	96.71 [96.31, 97.11]
GraphINR-IDS	99.07 [98.98, 99.15]	99.69 [99.61, 99.75]

All values in percent

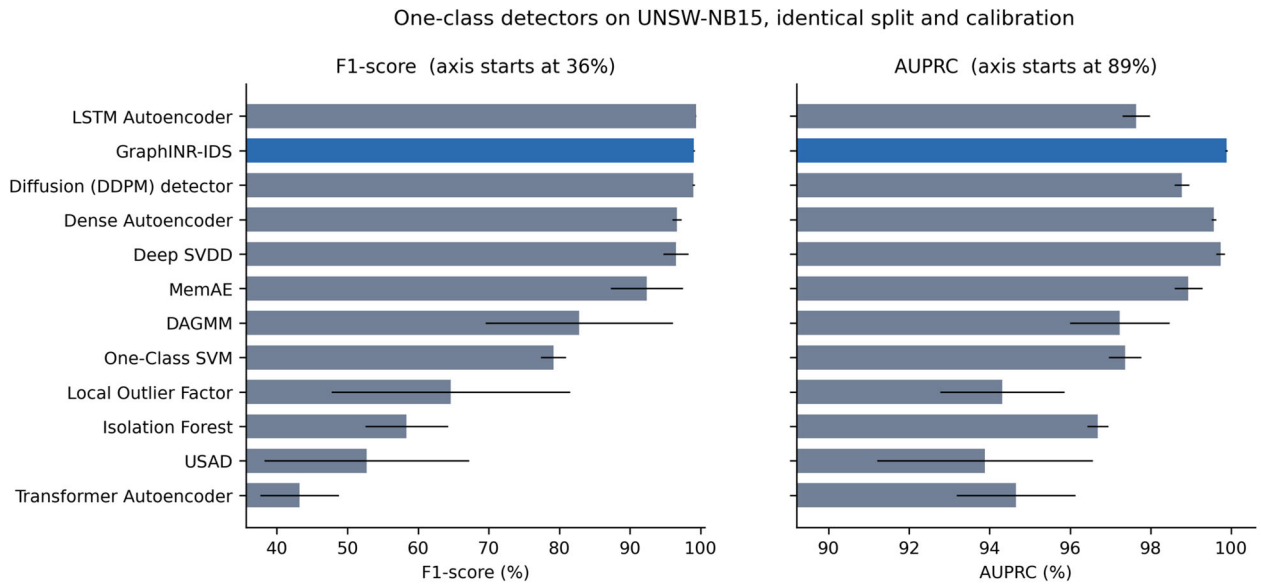


Fig. 2 One-class detectors on UNSW-NB15 under the random split. Bars are means over seeds and whiskers standard deviations. The four strongest methods are separated by less than the width of their own variability

Window length

An LSTM autoencoder was retrained with windows of 8, 16, 32 and 64 flows, while GraphINR-IDS continued to score each flow independently. The comparator matched the proposed model closely at 8 and 16 flows, at 99.34 % and 99.32 % F1. It then fell to 44.42 % at 32 and 32.32 % at 64, declines of roughly 55 and 67 percentage points, with AUROC following from 97.98 % to 87.55 %. The mechanism is visible in the architecture. The encoder compresses each window into a fixed 64-dimensional state before reconstructing it, so a longer window asks a fixed-capacity bottleneck to carry more information, and reconstruction degrades for benign and attack windows alike. This establishes that the tested fixed-capacity sequence model is strongly sensitive to a choice made before learning, and that a pointwise field has no equivalent choice to make. It does not establish that every discrete detector behaves this way, since a comparator with capacity scaled to window length might not.

Timestamp quality

With the model held fixed, jittering test timestamps by up to a minute or quantising them into ten-minute bins changed detection by less than a tenth of a percentage point. The field therefore does not depend on fine temporal resolution. The coordinate is not ignored, however. Replacing every timestamp with a single constant drove the false-

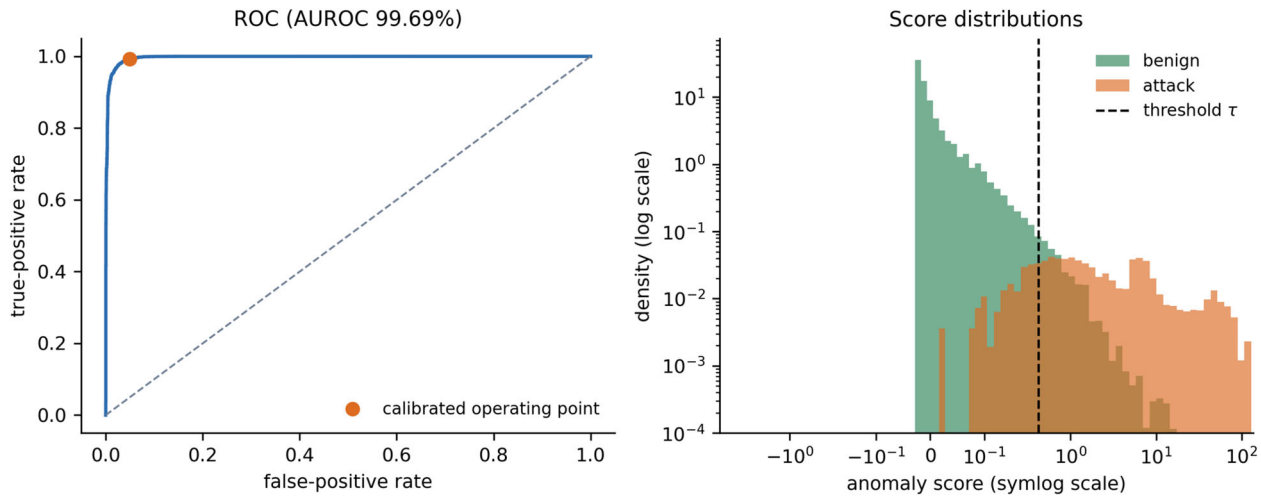


Fig. 3 Full model at seed 0. Left: ROC with the calibrated operating point marked. Right: benign and attack score distributions on a symlog horizontal axis and a log density axis, required because 97 % of the attack-score mass lies above τ but spread over nearly two decades

Table 9 Per-family detection under Protocol I with a single globally calibrated threshold. Because training uses benign flows only, every family listed is unseen during training, and the aggregate of these rates is the recall of Table 7

Attack family	Test flows	Detection rate (%)
Backdoor	244	100.00 ± 0.00
Worms	14	100.00 ± 0.00
Generic	17,356	100.00 ± 0.01
DoS	969	99.94 ± 0.09
Exploits	3,107	99.59 ± 0.20
Reconnaissance	973	99.51 ± 0.38
Shellcode	107	99.30 ± 1.09
Analysis	245	97.27 ± 1.48
Fuzzers	1,981	93.01 ± 2.32
Aggregate (= recall)	25,000	99.34 ± 0.23

Counts are means over seeds; each seed redraws the attack sample

alarm rate to 100 %. A constant coordinate is itself a value the field never saw while fitting, so every flow presented at it reconstructs badly. The accompanying 89.29 % F1 is the degenerate value obtained when everything is flagged. With 6,000 benign and 25,000 attack test flows, precision is $25{,}000/31{,}000 = 80.65\,\%$ at 100 % recall. Permuting timestamps among flows preserves the marginal distribution but destroys the association between a flow and its time. This cost about a point of F1 and raised false alarms to roughly 16 %. The field has learned a real but coarse dependence on time. The coordinate must be plausible, but need not be precise.

Removing the coordinate entirely

Retraining with the time coordinate held constant throughout reached 98.86 ± 0.30 % F1 and 99.58 ± 0.07 % AUROC, against 99.13 ± 0.07 % and 99.69 ± 0.04 % with it active. The gap of 0.27 percentage points is small but consistent in sign across all three seeds. This agrees with the degradation experiments. The coordinate does measurable work, and that work is modest on this benchmark.

Sampling density

Thinning the test stream from 100 % down to 10 % moved F1 for both detectors by less than a third of a percentage point, with neither trend monotone. This is a null result, and its cause lies in how the comparator is scored. Each flow is scored by the mean reconstruction error over all stride-one windows containing it, so its score is already largely

Table 10 Testing the continuity argument (Protocol I). The first block varies the window length of a discrete sequence detector; the second degrades the test-time timestamp with the model held fixed; the third retrains with the temporal coordinate held constant; the fourth thins the test stream and rescores the surviving flows. Three seeds

Experiment	Condition	F1	AUROC	FPR
Window length	8 flows	99.34 ± 0.06	97.98 ± 0.08	5.18 ± 0.64
	16 flows	99.32 ± 0.05	96.56 ± 0.37	5.32 ± 0.52
	32 flows	44.42 ± 3.20	91.08 ± 0.34	5.37 ± 0.54
	64 flows	32.32 ± 1.12	87.55 ± 0.32	5.01 ± 0.15
spread over window length		*67.02 pp*		
Timestamp degradation	exact timestamps	99.13 ± 0.07	99.69 ± 0.04	5.16 ± 0.26
	jitter $\sigma = 1$ s	99.13 ± 0.07	99.69 ± 0.04	5.16 ± 0.26
	jitter $\sigma = 10$ s	99.13 ± 0.07	99.69 ± 0.04	5.18 ± 0.29
	jitter $\sigma = 60$ s	99.13 ± 0.05	99.69 ± 0.04	5.29 ± 0.38
	quantised to 1 s	99.13 ± 0.07	99.69 ± 0.04	5.16 ± 0.26
	quantised to 60 s	99.13 ± 0.08	99.69 ± 0.04	5.13 ± 0.26
	quantised to 600 s	99.14 ± 0.03	99.69 ± 0.04	5.44 ± 0.41
	timestamp removed	89.29 ± 0.00	58.01 ± 25.29	100.00 ± 0.00
	timestamps permuted	98.03 ± 0.02	97.35 ± 0.44	16.32 ± 0.34
Temporal ablation	time coordinate active	99.13 ± 0.07	99.69 ± 0.04	5.16 ± 0.26
	time coordinate constant	98.86 ± 0.30	99.58 ± 0.07	5.18 ± 0.48
Stream thinning	100 % retained	99.14 ± 0.10 / 98.80 ± 0.80	99.71 ± 0.03 / 96.41 ± 0.50	5.16 ± 0.37 / 5.37 ± 0.78
	50 % retained	99.12 ± 0.15 / 98.84 ± 0.82	99.67 ± 0.05 / 96.57 ± 0.54	5.03 ± 0.40 / 4.96 ± 0.73
	25 % retained	99.22 ± 0.04 / 98.94 ± 0.71	99.72 ± 0.02 / 96.89 ± 0.56	4.80 ± 0.49 / 4.60 ± 0.49
	10 % retained	99.09 ± 0.07 / 98.71 ± 0.95	99.73 ± 0.15 / 95.85 ± 1.17	5.08 ± 1.04 / 5.83 ± 1.63
(values shown as GraphINR-IDS / windowed detector)				

All values in percent

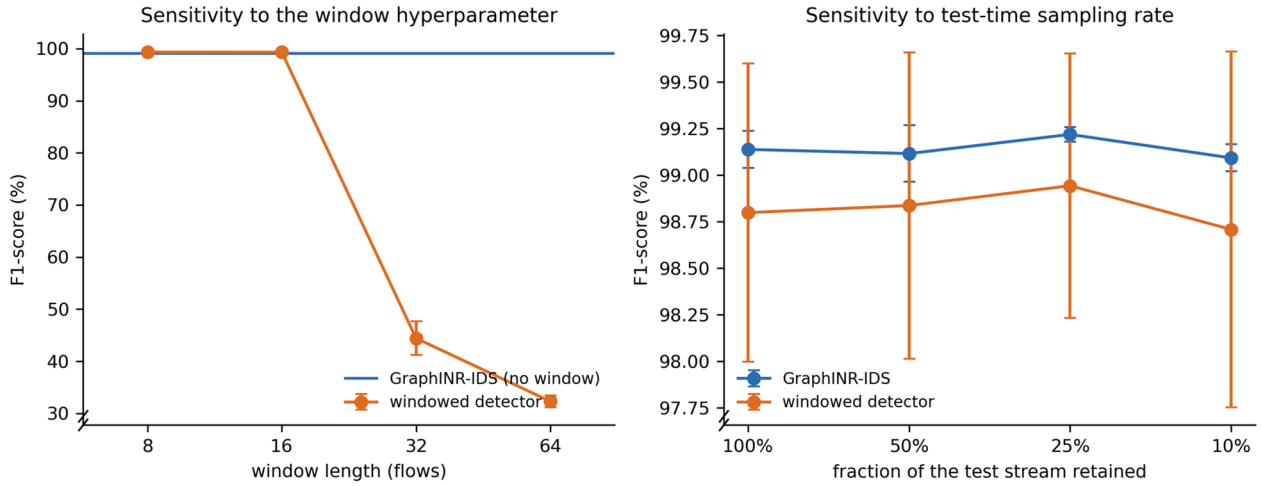


Fig. 4 Left: a windowed sequence detector as a function of its window length, against GraphINR-IDS, which has no such hyperparameter. Right: both detectors on progressively thinned test streams, scored on the surviving flows only. Both vertical axes are truncated below their lowest plotted value, as the break marker at each origin indicates; the right-hand panel spans roughly two percentage points

insensitive to window composition, which is the property the experiment was built to expose. Once scored this way, a windowed detector is closer to density-invariant than the argument for continuity assumes on this data. What separates the two is ranking quality rather than stability. GraphINR-IDS holds AUROC between 99.67 and 99.73 % throughout, while the comparator sits at 95.85–96.89 % at every retained fraction.

6.3 Architectural component contributions

The contribution of each architectural component was isolated through paired ablations under identical data partitions, seeds and calibration settings, so the rows of Table 11 are directly comparable.

Removing graph-conditioned modulation left F1 unchanged at 99.06 %, a difference far inside the seed-to-seed spread, while removing 1.21 M parameters. Removing hierarchical frequency decomposition cost about a third of a percentage point. Its effect size is large relative to the between-seed variance but small in absolute terms, and the paired test does not reach significance at five seeds. Removing both left a 0.38 M-parameter network scoring 98.65 ± 0.34 % F1, within half a point of the full 1.98 M-parameter model.

The SIREN reconstruction network therefore provides essentially all of the measured detection capability on these benchmarks, and neither refinement contributes materially to it. Graph-conditioned modulation was designed for graphs that evolve between fitting and inference, a condition neither benchmark supplies, so this result bounds what has been

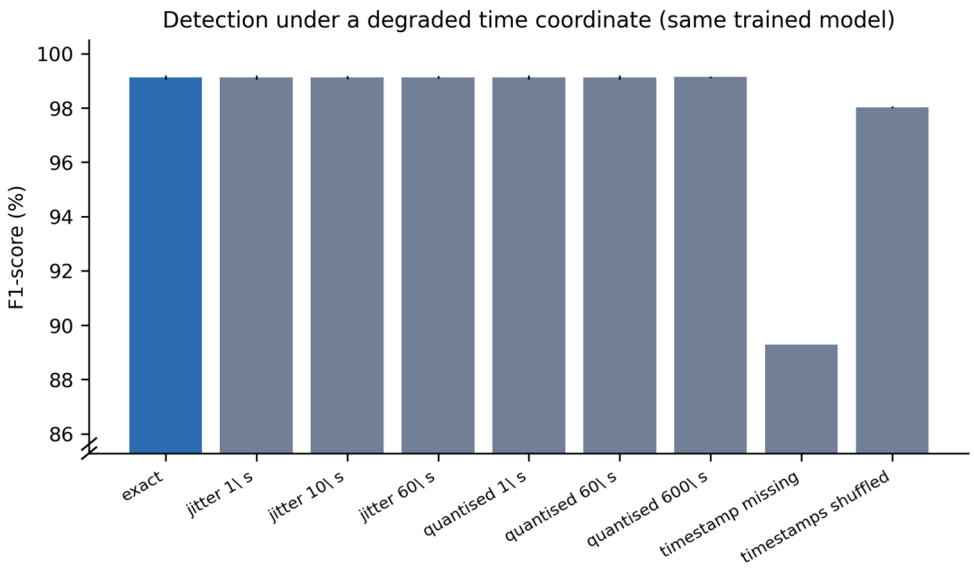


Fig. 5 Detection under a degraded test-time timestamp, same fitted model throughout. The vertical axis is truncated below its lowest plotted value, as the break marker at the origin indicates

Table 11 Component ablation under Protocol I. All configurations share the split, the calibration procedure and the seeds, so rows are paired. The right-hand block gives the paired difference from the full model in F1, with Cohen's d_z and a paired t-test; with five seeds the effect size is the more informative of the two

Configuration	Params	Recall	F1	AUROC	AUPRC	FPR	ΔF1 (pp)
GraphINR-IDS (full)	1,983,212	99.34 ± 0.23	99.06 ± 0.12	99.66 ± 0.04	99.88 ± 0.02	5.13 ± 0.18	—
w/o GCM	777,964	99.38 ± 0.22	99.06 ± 0.08	99.61 ± 0.04	99.87 ± 0.02	5.26 ± 0.36	+0.00
w/o HFD	1,585,700	98.73 ± 0.65	98.75 ± 0.30	99.51 ± 0.10	99.84 ± 0.05	5.13 ± 0.37	−0.31
backbone only (w/o GCM, w/o HFD)	380,452	98.52 ± 0.73	98.65 ± 0.34	99.48 ± 0.07	99.84 ± 0.03	5.07 ± 0.39	−0.41

All values in percent except parameter counts. A positive ΔF1 means the ablated variant scores *higher* than the full model. Paired effect size and t-test against the full model — w/o GCM, $d_z = 0.06$, $p = 0.907$; w/o HFD, $d_z = -1.53$, $p = 0.027$; backbone only (w/o GCM, w/o HFD), $d_z = -1.81$, $p = 0.016$

tested rather than refuting the design. It is retained as an option, with no positive claim attached. Hierarchical frequency decomposition retains a small advantage whose sign is stable across seeds.

6.3.1 Diagnostics against identity behaviour

The observed metadata is an input, so a network that copied it would reconstruct benign and attack traffic equally well and detect nothing. No term in the fitting objective forbids this. Table 12 therefore measures the fitted model against the behaviour an identity map would produce.

The fitted model is far from identity behaviour on every diagnostic. The mean diagonal of the input–output Jacobian is 0.243, where an identity map gives 1.0. The off-diagonal row mass is 1.745, where an identity map gives 0, so each reconstructed feature draws mostly on other features rather than on itself. Reconstruction error is 280.8 times larger on attack flows than on benign flows, and swapping a flow's graph-temporal coordinate inflates its error by a factor of 49.89. These are properties of the fitted network rather than guarantees from the objective, and they would require rechecking on any new deployment.

6.3.2 Spectral coordinate and endpoint novelty

The spectral coordinate was evaluated on two axes. The first is its measured contribution to detection. The second is the detector's behaviour when previously unseen endpoints appear. Table 13 reports both, beginning with a census of topological novelty, since the second measurement is interpretable only against the amount of novelty the test partition contains.

Under the random split there is almost no novelty. The UNSW-NB15 testbed has fewer than fifty endpoints, and the test partition contains no unseen nodes at all and a handful of unseen endpoint pairs covering four flows out of nearly thirty-one thousand. The chronological split differs, leaving five unseen endpoints and twenty-three unseen pairs covering 8,349 of 60,000 test flows. No test sample is discarded under either.

Removing the spectral encoding entirely and retraining cost about 1.15 percentage points of F1, at 99.13 % with the coordinate and 97.98 % without, over three seeds. Set against the refinements, the ordering is clear. The spectral coordinate costs 1.15 percentage points, hierarchical frequency decomposition 0.31, and graph-conditioned modulation nothing measurable. The two deltas come from separate ablations, the first over three seeds against a 99.13 % baseline and the second over five against 99.06 %, so the ordering is robust but the difference between them is not exact. The contribution is not an artefact of novelty acting as a shortcut. A control detector that flags a flow purely because its endpoint pair was unseen recovers 5.08 % of attacks against the full detector's 99.34 % recall, and restricting evaluation to flows on previ-

Table 12 Identity-mapping diagnostics for the full model under Protocol I, averaged over seeds. The right-hand column gives the value each quantity would take if the network had collapsed to $\hat{\mathbf{m}} = \mathbf{m}$

Diagnostic	Measured	Identity map gives
Mean reconstruction error, benign flows	1.255e-05	0
Mean reconstruction error, attack flows	3.434e-03	0
Ratio, attack / benign	280.8×	1.0
Mean $\lvert\partial \hat{m}_k / \partial m_k\rvert$ (Jacobian diagonal)	0.243	1.0
Mean off-diagonal Jacobian row mass	1.745	0.0
Error multiplier when coordinates are swapped	49.89×	1.00×
Recovery of masked features (50 % masked)	60.8 %	0 %
Median per-feature R^2, benign flows	0.999	1.000

Table 13 Topological novelty in the test partitions, and what the topology coordinate contributes. No test sample is discarded under any protocol. Three seeds; the census column is reported for seed 0, whose draw places four flows on an unseen edge

Quantity	Protocol I	Protocol II
Training-graph nodes	48	43
Training-graph edges	218	217
Test nodes	44	45
Test nodes unseen in training	0	5
Test edges unseen in training	4	23
Test flows on an unseen edge	4	8,349
Test samples discarded	0	0
Contribution of the topology coordinate (Protocol I, F1 in %)		
All test flows	99.13	
Flows on previously seen edges only	99.09	
Topology-novelty-only detector (recall)	5.08	
Spectral coordinate ablated, retrained	97.98 ± 0.37	
Endpoints forced out-of-vocabulary (Protocol I)		
10 % of endpoints hidden	F1 97.73 ± 0.37	FPR 17.64 ± 3.22
25 % of endpoints hidden	F1 94.04 ± 0.39	FPR 51.39 ± 3.29
50 % of endpoints hidden	F1 91.39 ± 0.67	FPR 77.99 ± 6.92
100 % of endpoints hidden	F1 89.29 ± 0.00	FPR 100.00 ± 0.00

ously seen edges leaves performance unchanged at 99.13 % against 99.09 % F1.

The cost appears when endpoints churn. Forcing a growing fraction of test endpoints onto the reserved node raised the false-alarm rate from about 5 % to 17.64 % at a tenth, 51.39 % at a quarter, 77.99 % at a half, and 100 % when every endpoint was unknown. The mechanism is the one the constant-timestamp experiment already exposed. The zero coordinate is a value the field never saw while fitting, so it is itself off-manifold. The operationally decisive comparison is with the ablated variant, which scores 97.98 % F1 at its nominal false-alarm rate. The full model drops below that as soon as a tenth of endpoints are unknown, at 97.73 % F1 and a 17.64 % false-positive rate. For a network with substantial endpoint churn the coordinate as specified is a liability rather than an asset.

Table 14 Protocol II: UNSW-NB15 cross-session, fitted on campaign 1 and tested on campaign 2, a gap of 26 days. Fitting and calibration use benign traffic that strictly precedes the test partition. Values are the two individual runs, printed as run 1 / run 2, not mean ± SD: a standard deviation estimated from two observations is not interpretable, and reporting one would invite a precision the design does not support. No best-in-column value is emphasised, and no ordering is claimed between methods whose two runs overlap. The Transformer Autoencoder and the Diffusion (DDPM) detector ran once, so a single value is shown for them

Method	Acc.	Prec.	Recall	F1	AUROC	AUPRC	FPR
USAD	74.50 / 78.30	84.61 / 84.49	75.49 / 82.61	79.79 / 83.54	79.25 / 78.83	86.27 / 85.21	27.46 / 30.33
DAGMM	78.03 / 80.45	83.43 / 84.00	83.66 / 87.31	83.54 / 85.62	84.38 / 77.02	91.29 / 83.22	33.23 / 33.26
Isolation Forest	83.91 / 81.42	85.01 / 84.81	92.11 / 87.86	88.42 / 86.31	89.12 / 88.55	94.14 / 93.75	32.50 / 31.47
Local Outlier Factor	84.00 / 81.62	82.86 / 82.24	95.81 / 92.39	88.87 / 87.02	71.37 / 70.98	74.52 / 74.11	39.63 / 39.91
MemAE	82.55 / 84.48	84.01 / 84.79	91.18 / 93.50	87.45 / 88.93	72.42 / 75.02	71.81 / 74.91	34.72 / 33.55
Transformer Autoencoder	83.97	85.10	92.08	88.45	83.65	88.87	32.25
Dense Autoencoder	87.08 / 84.97	84.84 / 85.09	98.16 / 93.92	91.02 / 89.28	82.94 / 74.65	87.75 / 76.12	35.08 / 32.92
Diffusion (DDPM) detector	86.16	85.13	96.02	90.25	79.56	81.90	33.55
Deep SVDD	87.06 / 85.32	84.61 / 85.00	98.50 / 94.69	91.03 / 89.58	91.38 / 73.12	93.97 / 73.03	35.83 / 33.42
LSTM Autoencoder	86.62 / 87.56	84.06 / 84.87	98.63 / 98.99	90.77 / 91.39	72.59 / 72.37	73.25 / 71.84	37.40 / 35.30
One-Class SVM	87.09 / 87.68	85.29 / 85.71	97.44 / 97.84	90.96 / 91.37	80.14 / 80.66	84.34 / 84.27	33.62 / 32.62
GraphINR-IDS	88.84 / 89.28	85.91 / 86.32	99.60 / 99.73	92.25 / 92.54	76.92 / 90.29	76.07 / 93.90	32.67 / 31.62

All values in percent

6.4 Calibration stability under chronological shift

Calibration stability was assessed by transferring a fitted model and its threshold across capture campaigns. Protocol II fits on the first UNSW-NB15 campaign and tests on the second. Every test timestamp therefore lies outside the fitting window, the benign traffic and attack mix have both moved, and the test partition contains endpoints the fitting graph never saw. Table 14 reports the outcome.

Two columns vary too much between runs to support a ranking. GraphINR-IDS records AUROC 76.92 / 90.29 % and AUPRC 76.07 / 93.90 %, a spread of more than thirteen points across a single reseeding. Discussion is therefore confined to F1 and to the realised false-positive rate, which are stable.

The transferred thresholds produced realised false-positive rates of 27 to 40 % across all evaluated detectors, against about 5 % under the random split. The resulting operating points are unsuitable. The threshold rule was unchanged throughout and remained the 95th percentile of benign calibration scores, so a threshold fitted on benign traffic from one capture period does not transfer to benign traffic from the next. The effect reaches the classical detectors, the deep ones and the proposed model alike, which points to a property of the traffic rather than of any method. A random split conceals it entirely by drawing calibration and test benign flows from the same period.

The threshold-dependent and threshold-free metrics also part company here, which is what a collapsed operating point looks like. GraphINR-IDS attains the highest F1 of the twelve, yet it does not lead on AUROC. Isolation Forest sits 5.04 percentage points lower on F1 and behaves quite differently at the transferred threshold. The recall and false-positive columns give the mechanism. The reconstruction-based detectors flag almost everything, which favours recall over precision, while Isolation Forest stays conservative and detects less. Neither constitutes a usable operating point. Because the AUROC column varies too much between runs to establish which method orders the drifted traffic best, this disagreement is reported as a symptom of the calibration failure rather than as an ordering of methods.

Table 15 Label-free deployment policies under Protocol II, where the test traffic post-dates the fitting window by 26 days. *Clipped* bounds the normalised time coordinate to the fitted range instead of letting it run to $t \approx 830$. *Burn-in* re-derives the threshold, and optionally the residual statistics, from the earliest B benign flows of the deployment period; those flows are excluded from scoring. All policies are label-free

Time coord.	Calibration	Recall	F1	AUROC	FPR
unclipped	carried over from fitting period	99.68 ± 0.07	92.31 ± 0.20	81.99 ± 7.24	32.55 ± 0.88
	threshold, burn-in $B = 500$	26.67 ± 35.66	33.27 ± 39.98	81.98 ± 7.22	7.86 ± 3.81
	threshold + statistics, $B = 500$	86.35 ± 0.32	88.72 ± 0.32	91.81 ± 1.85	17.05 ± 0.63
	threshold, burn-in $B = 1,000$	26.07 ± 34.94	32.84 ± 39.68	82.19 ± 7.08	7.26 ± 3.64
	threshold + statistics, $B = 1,000$	84.95 ± 0.55	88.20 ± 0.25	89.68 ± 2.87	16.16 ± 0.37
	threshold, burn-in $B = 2,000$	24.30 ± 32.73	31.51 ± 38.53	82.41 ± 7.08	6.75 ± 1.44
	threshold + statistics, $B = 2,000$	81.28 ± 5.16	86.38 ± 3.18	89.85 ± 2.74	15.07 ± 1.19
	threshold, burn-in $B = 5,000$	21.17 ± 32.80	27.28 ± 39.93	84.98 ± 5.70	3.05 ± 2.49
	threshold + statistics, $B = 5,000$	82.46 ± 1.68	87.87 ± 1.57	91.29 ± 3.78	13.90 ± 3.37
	threshold, burn-in $B = 10,000$	22.86 ± 33.37	29.53 ± 39.78	85.08 ± 4.99	5.06 ± 3.34
	threshold + statistics, $B = 10,000$	82.62 ± 1.26	88.08 ± 1.40	90.72 ± 4.10	19.94 ± 5.59
clipped	carried over from fitting period	99.72 ± 0.15	92.31 ± 0.23	82.75 ± 7.42	32.67 ± 0.82
	threshold, burn-in $B = 500$	24.41 ± 36.97	30.02 ± 42.28	82.74 ± 7.41	5.87 ± 1.72
	threshold + statistics, $B = 500$	86.10 ± 0.49	88.85 ± 0.24	91.29 ± 3.17	15.81 ± 0.44
	threshold, burn-in $B = 1,000$	24.34 ± 36.88	30.17 ± 42.48	82.97 ± 7.27	3.85 ± 1.90
	threshold + statistics, $B = 1,000$	84.94 ± 0.55	88.26 ± 0.27	89.86 ± 0.78	15.89 ± 0.24
	threshold, burn-in $B = 2,000$	24.34 ± 36.88	30.20 ± 42.52	83.23 ± 7.27	3.85 ± 1.94
	threshold + statistics, $B = 2,000$	83.86 ± 1.35	87.89 ± 0.74	89.75 ± 0.65	15.46 ± 0.53
	threshold, burn-in $B = 5,000$	24.34 ± 36.88	30.26 ± 42.60	85.72 ± 5.88	3.66 ± 2.44
	threshold + statistics, $B = 5,000$	82.82 ± 2.47	87.64 ± 1.73	90.90 ± 2.44	16.39 ± 1.57
	threshold, burn-in $B = 10,000$	24.35 ± 36.89	30.29 ± 42.63	86.04 ± 5.31	4.71 ± 3.89
	threshold + statistics, $B = 10,000$	83.49 ± 1.47	88.23 ± 1.29	91.41 ± 2.81	22.98 ± 3.62

All values in percent; three seeds

6.4.1 Origin of the drift and label-free recalibration

The temporal encoding must be ruled out before attributing the degradation to distribution shift, because it is demonstrably broken outside its fitted window. Normalising against a 2,781 s fitting window places Protocol II test flows at $t \approx 830$. At 10 kHz the phase then exceeds 5×10^7 radians, where one float32 unit in the last place of t shifts it by more than half a cycle. Measured directly, every channel of the temporal encoding has standard deviation $1/\sqrt{2}$ on the test partition, exactly that of a sinusoid at uniformly random phase, against 0.30–0.74 on the fitting partition. The coordinate is not merely aliased there. It carries no information at all.

That degeneracy is nonetheless not the cause. Clipping the coordinate to its fitted range changed the false-positive rate by about a tenth of a point and AUROC by under a point (Table 15). This is consistent with the temporal ablation, since on this data the coordinate contributes little either way, so degrading it costs little and repairing it gains little. The degeneracy would matter on data where time carries more signal.

The benign traffic distribution itself has moved. The two label-free recalibration policies in Table 15 behave very differently. Both draw on the earliest B benign flows of the deployment period, excluded from scoring so that no flow is judged against a threshold derived from itself.

Re-deriving only the threshold restores the nominal false-positive rate and makes detection erratic. Across three seeds at a 2,000-flow burn-in, recall falls to 0.3 %, 11.0 % and 61.6 %, and the standard deviation of F1 exceeds its mean. The drifted benign score distribution now overlaps the attack distribution, so the 95th percentile of current benign scores lands somewhere among the attack scores. Where it lands depends on whichever tail happens to be heavier in a given run.

Re-deriving the per-feature residual statistics alongside the threshold behaves differently. Recall recovers to 81.28–86.35 % at a false-positive rate between 13.90 and 19.94 %, and the standard deviation of recall across seeds falls from more than thirty points to between 0.32 and 5.16. AUROC also improves on the carried-over configuration, from 81.99 to 90.67 on average. Restandardising against current benign residuals therefore does not merely move the operating point. It restores part of the score ordering, which is consistent with removing a component of drift common to all flows in the new period. The result still falls short of the Protocol I operating point, which bounds what an unlabelled burn-in achieves. The practical implication is not specific to this model. A reconstruction detector deployed against traffic post-dating its fitting window should refresh what it standardises against, not only where it cuts, and a burn-in of a few hundred to a few thousand benign flows suffices.

6.5 Cross-Network transfer and deployment conditions

Transfer to a second network was evaluated on CIC-IDS2017, fitting on its single attack-free day and testing on the remaining four, with all fourteen attack classes unseen at fitting time. Table 16 reports the comparison, which covers nine of the eleven baselines. The Transformer Autoencoder and the diffusion detector are not included.

GraphINR-IDS is mid-field here, at 61.78 % F1 against 67.84 % for the LSTM Autoencoder, which leads on both F1 and AUROC. Each method was run once under this protocol, so that lead rests on a single run. The margins are

Table 16 Protocol III: CIC-IDS2017 cross-day, fitted on Monday, the only attack-free day, and tested on Tuesday–Friday, so all fourteen attack classes are unseen at fitting time. Each method was run once, so no dispersion is reported and small differences should not be read as an ordering. Protocol III was run for 9 of the eleven baselines. The graph and time coordinates are inert here because the redistributed CIC-IDS2017 features omit endpoint identifiers and timestamps (Section 5.2), so this protocol tests the reconstruction principle rather than the full model

Method	Acc.	Prec.	Recall	F1	AUROC	AUPRC	FPR
Isolation Forest	52.11	93.75	30.18	45.65	66.72	81.75	4.03
Deep SVDD	59.62	95.68	41.30	57.69	62.25	80.71	3.73
USAD	59.99	94.82	42.30	58.50	59.20	80.92	4.62
DAGMM	61.07	95.00	43.91	60.06	58.91	81.03	4.62
MemAE	61.51	95.72	44.25	60.52	77.31	86.87	3.96
Dense Autoencoder	61.79	95.97	44.55	60.85	67.03	83.08	3.74
Local Outlier Factor	61.00	90.36	46.47	61.37	80.30	84.36	9.92
One-Class SVM	62.91	94.70	46.99	62.82	77.84	88.82	5.27
LSTM Autoencoder	66.80	95.76	52.52	67.84	91.25	94.50	4.65
GraphINR-IDS	62.40	95.80	45.59	61.78	77.37	88.27	4.00

All values in percent; one run per method

Table 17 Per-class detection on CIC-IDS2017 under Protocol III, at the benign-calibrated threshold. All fourteen classes are unseen at fitting time. One run, so no dispersion is reported. The split is sharp: volumetric denial-of-service classes are detected at between 58.25 and 94.41%, whereas the enumerative classes — port scanning and the two brute-force families — are detected at essentially zero, because what makes them attacks is the pattern across flows

Attack class	Test flows	Detection rate (%)
Heartbleed	1	100.00
Infiltration	2	100.00
DoS Slowhttptest	429	94.41
DoS GoldenEye	792	67.55
DoS Hulk	16,551	67.48
DDoS	9,103	64.06
DoS slowloris	400	58.25
Web Attack – XSS	35	5.71
Web Attack – Brute Force	99	2.02
Bot	151	1.99
PortScan	11,419	0.48
FTP-Patator	567	0.00
SSH-Patator	449	0.00
Web Attack – SQL Injection	2	0.00

Heartbleed and Infiltration have one and two test flows respectively, so their rates carry no weight

6.06 percentage points of F1 and 13.88 of AUROC. Three observations follow. Calibration transfers cleanly this time, with realised false-positive rates near the 5 % target for every detector but Local Outlier Factor, which reaches 9.92 %. This locates the Protocol II failure in the 26-day gap rather than in a change of network or attack mix. Absolute difficulty is far higher, since no detector exceeds 68 % F1 and recall stays below 55 % throughout. The ranking also changes. The redistributed feature set carries neither endpoint identifiers nor timestamps, so both coordinates that distinguish the method are inert and the model reduces to a SIREN autoencoder with top-κ residual scoring. This result bounds what Protocols I and II establish. The reconstruction machinery alone, without graph or time, is competitive but not distinctive.

The breakdown by family in Table 17 locates the difficulty. Denial-of-service families are detected at between 58.25 and 94.41 %, because flooding pushes volume and rate features far outside the benign envelope. Port scanning and the FTP and SSH brute-force families are detected at essentially zero. Each of their individual flows is a short, well-formed connection indistinguishable at the metadata level from a benign short connection, and what makes them attacks is the pattern across flows. A per-flow detector cannot see that pattern by construction. This is the clearest case in these results for the optional rolling-window aggregation, and it is a limit on per-flow reconstruction scoring generally rather than on this model in particular.

6.5.1 Detection under encryption-observable feature sets

Only 24 of the 36 benchmark features survive transport encryption, and 31 survive when only the payload is encrypted. Table 18 reports detection under each nested set, and Fig. 6 plots it.

The result is nearly flat. Discarding the twelve features that need a readable TCP header or application-layer inspection costs 0.20 percentage points of F1 and 0.03 points of AUROC, both inside the seed-to-seed spread. The volumetric, timing and connection-context features of Tier A therefore carry essentially all the detection signal on this benchmark, and the features added by Tiers B and C are largely redundant with them. This is favourable for the encrypted setting the paper targets. It is a property of this benchmark's feature redundancy rather than a general guarantee. A network whose attacks manifest specifically in TCP-window or HTTP-layer behaviour would not be well served by Tier A alone, and the nine families of UNSW-NB15 do not exercise that case.

6.5.2 Sensitivity to network and host conditions

UNSW-NB15 was captured under one set of network conditions, so the effect of congestion, loss, jitter, shaping and host load cannot be observed directly. It is instead studied by perturbing feature vectors according to an explicit model of each condition, with the fitted model and calibrated threshold held fixed. Seven conditions are applied at four intensities each, and Table 19 reports the outcome. Delay jitter is applied at relative standard deviation 0.1–1.0, congestion stretches duration and round-trip time by 10–100 %, and packet loss ranges over 1–10 %. Traffic shaping caps throughput from

Table 18 Detection when only the features observable under a given encryption regime are available (Protocol I). Tier A+B+C is the 36-feature benchmark set used in the literature; it is not observable in an encrypted deployment

Tier	Available under	Feat.	Recall	F1	AUROC	FPR
A — IP header and packet timing	TLS 1.3, QUIC, DTLS, IPsec, VPN	24	99.12 ± 0.22	98.94 ± 0.05	99.68 ± 0.03	5.19 ± 0.55
A+B — + readable TCP header	TLS 1.3 over TCP only	31	99.23 ± 0.49	98.97 ± 0.24	99.56 ± 0.20	5.44 ± 0.08
A+B+C — + application layer	none of the above (needs plaintext)	36	99.51 ± 0.28	99.14 ± 0.10	99.71 ± 0.03	5.16 ± 0.37

All values in percent. Tiers are nested; κ is scaled with the feature count as $\lceil d/4 \rceil$

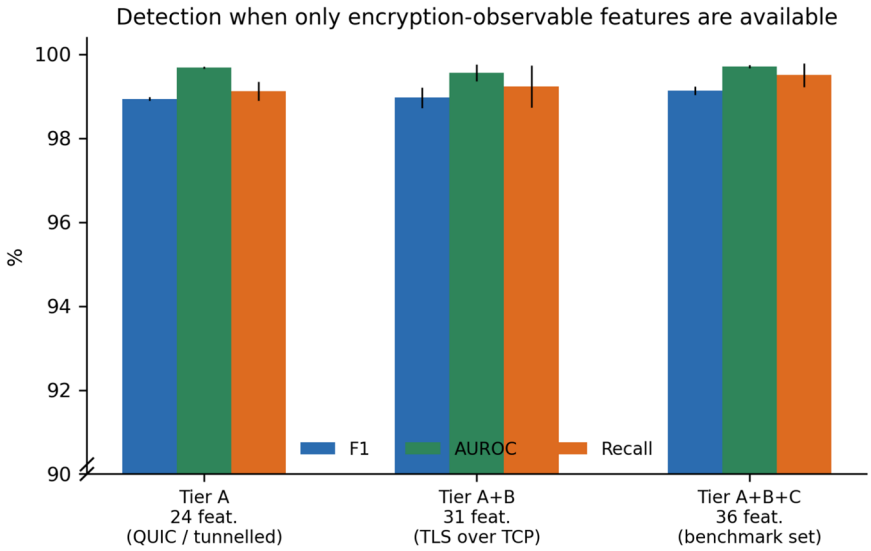


Fig. 6 Detection when only the features observable under a given encryption regime are available. The vertical axis is truncated at 90 %, as the break marker indicates; all plotted values lie between 98.94 and 99.71 %

the 95th down to the 50th percentile, host load adds 0.1–1.0 median-scaled offsets to response timings, routing changes shift TTL by ± 1 to ± 10, and MTU reduction scales packet counts up and mean sizes down by 10–100 %.

The conditions separate into two regimes. Perturbations acting on timing barely move the operating point. Congestion, jitter, shaping and host load all keep the false-alarm rate within about two points of its unperturbed value. Perturbations acting on the structure of the packet stream move it a great deal. A routing change shifting TTL by up to ten hops takes the false-alarm rate to 35.77 %, and an MTU reduction to 66.07 %. Packet loss sits between them at 16.93 % under 10 % loss, consistent with acting through the retransmission counters and the packet and byte totals rather than through timing. Re-deriving the threshold from perturbed benign traffic restores the false-alarm rate to within 0.1 points of its unperturbed value under every condition. This is the opposite of the chronological-drift behaviour, and the contrast is informative. These perturbations shift the benign score distribution without collapsing its separation from the attack distribution. These are models of each condition's effect on the feature vector rather than measurements from traces captured under it. They bound sensitivity, but say nothing about how large such perturbations are in a given network.

6.5.3 Scoring sensitivity and computational cost

The scoring stage is insensitive to its remaining free choices (Table 20). Varying the aggregation width from 3 to 36 moves F1 by 0.21 percentage points. Standardising by the variance rather than the standard deviation costs 0.10. Enabling the optional rolling-window aggregation raises recall to 100 % while lowering both F1 and AUROC. The fixed value $\kappa = 9$ is not the maximum of the sweep, which is what one expects of a value chosen in advance rather than on test performance.

The spectral stage reaches 10^6 nodes in 103.86 s on a single commodity CPU host. At deployment scale the binding constraint is therefore how often topology change forces recomputation rather than the decomposition itself. Table 21 gives measured throughput per configuration.

Table 19 Sensitivity to modelled network and host conditions (Protocol I). Perturbations are applied to raw feature values with the trained model and the calibrated threshold held fixed. FPR is measured with only benign flows perturbed; F1 with all test traffic perturbed. The last column re-derives the threshold on a perturbed benign calibration set at the strongest level

Condition	FPR (%) at increasing intensity				F1 (%) at level 4	FPR after recal.
	1	2	3	4		
unperturbed	5.16				99.13	—
Delay jitter	5.29	5.46	5.80	7.18	99.03	5.08
Congestion	5.13	5.29	5.51	6.05	99.17	5.14
Packet loss	5.70	7.13	9.24	16.93	97.77	5.09
Traffic shaping	5.15	5.14	5.20	5.21	99.14	5.17
Host processing load	5.16	5.16	5.16	5.16	99.13	5.16
Routing change (TTL shift)	5.24	5.33	6.84	35.77	95.72	5.12
MTU reduction	7.39	21.84	45.96	66.07	92.60	5.18

Intensity levels per condition are given in Section 6.5.2. These are models of each condition's effect on the feature vector, not measurements from traces captured under it

Table 20 Sensitivity of the scoring stage under Protocol I. κ is fixed *a priori* at 9 and is not selected; the sweep is reported as a robustness statement. Two seeds. The variance row compares the two denominators admitted by (4)

Setting	Value	Recall	F1	AUROC
Top-κ aggregation	$\kappa = 1$	98.67 ± 0.36	98.69 ± 0.20	99.58 ± 0.06
	$\kappa = 3$	99.21 ± 0.42	98.97 ± 0.18	99.66 ± 0.04
	$\kappa = 5$	99.38 ± 0.35	99.06 ± 0.12	99.69 ± 0.04
	$\kappa = 9$*	99.51 ± 0.28	99.14 ± 0.10	99.71 ± 0.03
	$\kappa = 12$	99.53 ± 0.27	99.16 ± 0.09	99.72 ± 0.03
	$\kappa = 18$	99.57 ± 0.23	99.17 ± 0.07	99.73 ± 0.03
	$\kappa = 27$	99.59 ± 0.21	99.18 ± 0.06	99.73 ± 0.03
	$\kappa = 36$	99.58 ± 0.23	99.18 ± 0.07	99.73 ± 0.03
Residual denominator	standard deviation*	99.51 ± 0.28	99.14 ± 0.10	99.71 ± 0.03
	variance	99.32 ± 0.51	99.04 ± 0.24	99.68 ± 0.09
Target FPR α	$\alpha = 0.01$	92.81 ± 0.68	96.16 ± 0.36	99.71 ± 0.03
	$\alpha = 0.02$	96.69 ± 0.26	98.06 ± 0.13	99.71 ± 0.03
	$\alpha = 0.05$*	99.51 ± 0.28	99.14 ± 0.10	99.71 ± 0.03
	$\alpha = 0.1$	99.97 ± 0.01	98.79 ± 0.01	99.71 ± 0.03
Score aggregation	per flow*	99.51 ± 0.28	99.14 ± 0.10	99.71 ± 0.03
	+ rolling 1/10/60 s	100.00 ± 0.00	98.87 ± 0.71	99.38 ± 0.03

* the reported configuration. All values in percent

7 Discussion

This section reads the results against the claims that motivated the design, separating what they establish from what they bound.

7.1 Continuous reconstruction and representation

The clearest benefit of a pointwise field is the absence of a decision. A windowed detector requires a window length before learning begins, and the tested comparator lost roughly 55 to 67 percentage points of F1 as that length grew from 8 to 64 flows. A field scored at a flow's own coordinate offers nothing to choose. This is the property that motivated the design and the one the evidence supports most directly.

Its limits are equally clear. The claim concerns a hyperparameter that need not be selected, not accuracy. On a random split the proposed model and the best windowed detector differ by less than the spread across seeds. The result also does not generalise to all discrete detectors, since the comparator's collapse traces to a fixed-capacity bottleneck that a differently scaled architecture might avoid. The compan-

Table 21 Cost of the one-off spectral stage as a function of graph size, and per-configuration inference cost. Graphs are generated with the mean degree measured on the UNSW-NB15 training graph (9.08). The eigen-solve extracts the $K = 32$ smallest non-trivial eigenpairs by Lanczos iteration at tolerance 10^{-3} and is performed once, before deployment

Nodes	Edges	Eigen-solve (s)	Peak memory (MiB)
100	338	0.02	0.2
500	1,809	0.08	0.7
1,000	3,867	0.11	1.3
5,000	21,822	0.36	6.6
10,000	44,515	0.95	13.1
50,000	224,251	3.14	65.3
100,000	449,926	7.21	130.5
500,000	2,262,549	57.39	652.4
1,000,000	4,532,915	103.86	1304.7
Configuration	Parameters	Flows/s	μs/flow
full	1,983,212	6,901	145
w/o GCM	777,964	12,279	81
w/o HFD	1,585,700	7,601	132
backbone only	380,452	16,835	59

Inference measured on the CPU host used for this revision; figures are comparative across configurations rather than deployment specifications

ion prediction failed outright. Thinning the test stream was expected to separate a pointwise detector from a windowed one, and did not, because scoring each flow over all windows containing it already confers most of the invariance at issue.

7.2 Topological context and endpoint novelty

The spectral coordinate earns its place on the available evidence. It improves F1 by 1.15 percentage points where it can be instantiated, which is more than either refinement contributes. The gain comes from genuine information about where a flow sits, not from novelty acting as a proxy for maliciousness. A detector flagging only unseen endpoint pairs recovers 5.08 % of attacks, and excluding those flows changes nothing.

The same coordinate is the model's most serious weakness in deployment. Endpoints outside the fitted graph receive a zero vector, which is not a neutral position but an off-manifold one. The false-alarm rate rises steeply as churn increases, passing the accuracy of a model with no topology at all once a tenth of endpoints are unknown. The defect lies in the encoding rather than the evaluation. A coordinate constructed inductively for unseen endpoints, for instance by aggregating the coordinates of the neighbours through which they communicate, would remove the discontinuity. Until that exists, the coordinate suits networks with stable endpoint populations, or deployments willing to refit the spectrum as hosts appear.

7.3 Calibration under distribution shift

The most transferable finding concerns neither architecture nor ranking. Across a 26-day gap the realised false-positive rate of every detector tested rose from about 5 % to between 27 and 40 % under an unchanged threshold rule. This is a property of the traffic, not of a method, and it is invisible to any evaluation that draws calibration and test data from the same period.

What recovers it is specific. Re-deriving the threshold alone restores the nominal rate while making detection erratic, because the drifted benign scores now overlap the attack scores and the quantile lands arbitrarily among them. Re-deriving the per-feature residual statistics as well recovers recall to 81.28–86.35 % and improves AUROC from 81.99 to 90.67. The correction therefore restores part of the ordering and not merely the operating point. The lesson generalises past this architecture. A reconstruction detector must refresh what it standardises against, not only where it cuts.

7.4 Interpretation of comparative performance

A random split of UNSW-NB15 places five one-class detectors within three percentage points of the leader and contains no unseen endpoints at all. It therefore discriminates on neither detection quality nor topological generalisation. Rankings drawn from it are fragile, and the bootstrap analysis shows how such a ranking can look solid while resting on a single fitted model. Flow-level intervals separate the three strongest methods, but retraining erases that separation.

Read across all three protocols, the evidence supports a bounded claim. No metric ranks GraphINR-IDS first under more than one protocol at a time. On CIC-IDS2017, where neither coordinate can be instantiated, it is mid-field and a windowed detector is stronger. The components added above the reconstruction network do not close that gap. The distinguishing properties are structural rather than numerical. The score is an absolute per-feature residual that decomposes into an attribution. A verdict is independent of how the surrounding stream is sampled. Topology is carried in the coordinate system rather than in a snapshot schedule.

7.5 Encrypted-Traffic applicability

Three distinct claims must be kept apart. The first is payload independence, which holds by construction, since nothing in the method reads packet contents. The second is feature observability, which the audit establishes and the restricted-feature experiment tests. The 24 features surviving every encryption regime carry essentially all the detection signal on this benchmark, at a cost of 0.20 percentage points of F1. The third is behaviour on genuinely encrypted traffic, and that has not been shown. Neither benchmark carries TLS or QUIC traffic, and encryption does more than remove features. It quantises packet sizes through record padding, shifts early flow timing through handshake cost, inflates jitter through cryptographic processing, and inside a tunnel changes what an address denotes. The perturbation study bounds sensitivity to changes of this kind but models them rather than measuring them. Injecting cryptographic processing overhead into flow traces, as Behdadnia et al. (2026) do, is the more faithful route and the natural next step.

8 Limitations

No transport-encrypted evaluation data

Neither benchmark contains TLS- or QUIC-encrypted traffic. Auditing feature observability and re-evaluating under the surviving sets bounds the claim but does not substitute for an encrypted trace.

Static and very small topology

The UNSW-NB15 graph has 48 endpoints and does not change between fitting and inference, so the spectral encoder is never exercised in the regime where the number of retained

eigenvectors is far smaller than the number of endpoints. CIC-IDS2017, as redistributed, carries no endpoint identifiers at all. An evaluation on a trace with an evolving IP graph, for which CESNET-TimeSeries24 (Koumar et al. 2025) is the obvious candidate, remains outstanding.

The refinements are not validated by these benchmarks
Graph-conditioned modulation contributes nothing measurable despite 1.21 M parameters, and hierarchical frequency decomposition contributes 0.31 percentage points. The former was designed for evolving graphs, a condition neither benchmark provides, so the null result bounds what has been tested rather than refuting the design.

The out-of-vocabulary coordinate degrades sharply
Mapping unknown endpoints to a zero spectral coordinate raises the false-alarm rate steeply as endpoint churn increases. This confines deployment to networks with stable endpoint populations, or to operators willing to refit the spectrum as hosts appear.

No formal protection against identity collapse
Neither data-dependent term in the fitting objective penalises an identity map, and the weight-magnitude penalty acts only indirectly. The diagnostics show that the fitted model is far from identity behaviour, but this is an empirical finding about these fits and would require re-checking elsewhere.

The temporal encoding does not extrapolate
Timestamps beyond the fitting window alias onto phases already seen, and outside that window the coordinate was measured to carry no information at all. Under chronological deployment the detector therefore relies on metadata and topology rather than on absolute time.

The window-length result is comparator-specific
The collapse observed as the window grew traces to a fixed-capacity bottleneck in the tested sequence model. A comparator whose capacity scaled with window length might not degrade in the same way, so the result does not establish a general property of discrete detectors.

Adversarial evasion and poisoning are out of scope
An evasion-aware adversary able to query the score is untested, as is poisoning of the calibration partition. Poisoning is the more pressing of the two. Both the threshold rule and label-free recalibration assume the burn-in is benign, and recalibration re-estimates residual statistics from the earliest flows of the deployment period without verifying them.

The perturbation study is a model, not a measurement
Network and host conditions were simulated by perturbing feature vectors. This bounds sensitivity but does not establish the magnitude of these effects in a real network.

9 Conclusion

GraphINR-IDS formulates benign network behaviour as a continuous graph-temporal reconstruction problem and derives anomaly decisions from standardised feature-wise residuals. Spectral coordinates of the graph Laplacian encode communication position, multi-frequency sinusoids encode capture time, and a SIREN network reconstructs each flow from its own coordinate. Fitting, residual standardisation and threshold calibration draw on benign flows alone, so the attack families met at test time are absent from fitting by construction.

The formulation is competitive where its coordinates can be instantiated. It attains the strongest threshold-free ranking of the twelve detectors compared under the random protocol, and it removes the sequence-window selection to which a fixed-capacity comparator proved highly sensitive. Component ablation places the spectral coordinate above both optional modules in measured contribution, at 1.15 percentage points of F1, with graph-conditioned modulation contributing nothing detectable on the static graphs available.

The most practically consequential result concerns calibration rather than architecture. Transferring a fitted model and its threshold across capture campaigns raised the realised false-positive rate for every detector evaluated. Recovery required re-estimating the feature-wise residual statistics rather than the threshold alone. A reconstruction detector must therefore refresh the reference distribution against which residuals are standardised, not merely the point at which it cuts.

Two constraints bound these conclusions. The zero-vector encoding of unseen endpoints becomes a liability as endpoint populations change, to the point where a model carrying no topological coordinate performs better once a modest fraction of endpoints is unfamiliar. And no benchmark used here carries operational TLS or QUIC traffic or an evolving communication graph, so applicability to encrypted deployment rests on a feature-observability audit rather than on direct measurement. Evaluation on traces that are genuinely encrypted and topologically dynamic is the evidence this formulation now requires.

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Data Availability The UNSW-NB15 dataset is publicly available at https://research.unsw.edu.au/projects/unsw-nb15-dataset and CIC-IDS2017 at https://www.unb.ca/cic/datasets/ids-2017.html. Neither is redistributed here, so no dataset licence is affected.
The implementation and the complete reproduction code accompany this submis sion at https://github.com/Deepan777/GraphINR-IDS. The repository contains the preprocessing code, the complete 36-feature list with the observability tiers of Table 3, seeded scripts that reconstruct every data partition from the original dataset archives, the random seeds, the model and baseline definitions, the training and evaluation scripts, the calibration procedure, pinned package versions, and scripts that regenerate every table, figure and in-text number in this paper.

Declarations

Competing Interests The authors declare no competing interests.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process During the preparation of this work the author(s) used Claude (Anthropic) in order to assist with reasoning and grammar editing. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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